Republic of Tunisia
The Ministry of Higher Education and Scientific Research

The University of Carthage

The Higher Institute of Information and Communication Technologies

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the

Bachelor Degree in Computer Science

Field of Study : Software Engineering and Information Systems

Design and Implementation of an AI-Based Network Intrusion Detection System

By

Amira BOUAFIF & Hajer JEMAA

Conducted within ITXTREE

Publicly defended on June 6, 2026 in front of the jury members:

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Academic Year: 2025-2026

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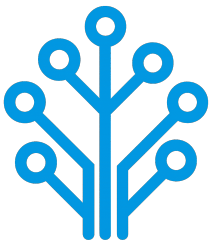

Authorization of graduation project report submission:

Professional Supervisor:	Academic Supervisor:
Issued on :	Issued on :
Signature:	Signature:

Dedications

I dedicate this End-of-Study project

To my dear parents

my dear father Elyes and my dear mother Sawsen

Who have supported and encouraged me every step of the way. This is for you, in honor of all the sacrifices you have made for me. I hope to have made you both proud and happy.

To my dear sisters and brothers

Oumayma, Eya, Yassmin, Mohamed and Yessin

Who have been a great support and a source of joy in my life. I am grateful for your love and encouragement throughout this journey.

To all my family and friends

Thank you all for your support and encouragement throughout all this time. I am truly blessed to have you in my life.

Amira BOUAFIF

Dedications

I dedicate this End-of-Study project

To my dear parents

To my dear sisters

To all my family and friends

Hajer JEMAA

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General Introduction

Chapter 1

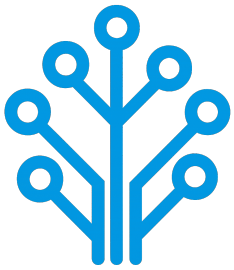
Project Context

1.1 Introduction

This chapter serves as an introduction to the project. We begin with a presentation of the company hosting the internship, a study of existing network intrusion detection systems. Then, we present the problem and the proposed solution. Finally, we end the chapter with a conclusion.

1.2 Presentation of the company

ITXTREE is an early-stage technology startup based in Sidi Rzig, Tunisia, focused on building a solid foundation for future digital solutions it is built on the core values of integrity, quality, curiosity and adaptability.



1.3 Study of the existing

Cybersecurity is a highly sensitive domain where the ability to react quickly can determine the impact of an attack. The earlier a threat is identified, the more effectively its consequences can be limited.

For this reason, Network Intrusion Detection Systems (NIDS) play a central role in monitoring and analyzing network traffic. The most widely used approach relies on:

- **Signature-based:** These systems analyze network traffic, often through deep packet inspection and compares it against a signature database. When a match is found an alert is triggered. Well-known tools such as Snort [1] and Suricata [2] are representative of this approach.

- **Anomaly-based:** Anomaly detection is learning normal behavior of the network through statistical analysis of traffic parameters. If there is a sudden increase of traffic on a port outside of normal thresholds, it might be an indication of a brand-new threat [1].

1.4 Criticism of the existing

Despite their widespread use, signature-based NIDS present several limitations:

- **High Maintenance and Reactivity:** Their effectiveness is essentially tied to the completeness and accuracy of their signature rule sets. As attack techniques continuously evolve, these systems require frequent updates to remain effective. This maintenance process is both time-consuming and reactive since new signatures can only be created after an attack has already been identified and analyzed.

- **Performance:** As the number of rules increases, the system must process a larger set of comparisons for each packet, which can lead to slower detection. In a context where reaction time is critical, this degradation directly affects the system's ability to respond effectively.

- **Rigidity and Evasion Vulnerability:** Attackers can exploit the rigidity of signature-based detection systems by introducing slight variations in attack patterns, allowing them to evade existing signatures while preserving the same malicious intent. Since many signature rules are publicly available or widely shared with the security community, attackers may analyze them to identify gaps and design variants that bypass detection mechanisms.

Overall, these systems rely on exact matching and predefined logics which limits their adaptability in the face of evolving and increasingly sophisticated threats.

1.5 Proposed solution

With the growing integration of Artificial Intelligence (AI) in security systems, new approaches aim to overcome the limits of traditional methods. To address these challenges, we propose an AI-driven NIDS capable of learning patterns from network traffic and classifying different types of network activity instead of relying solely on predefined rules.

1.6 Project Pipeline

A project pipeline is a structured sequence of stages that outlines the workflow and processes involved in the development and deployment of a project. It serves as a roadmap, guiding the team through various phases, from initial planning to final delivery.

Figure 1.1 illustrates the project pipeline for our network intrusion detection system.

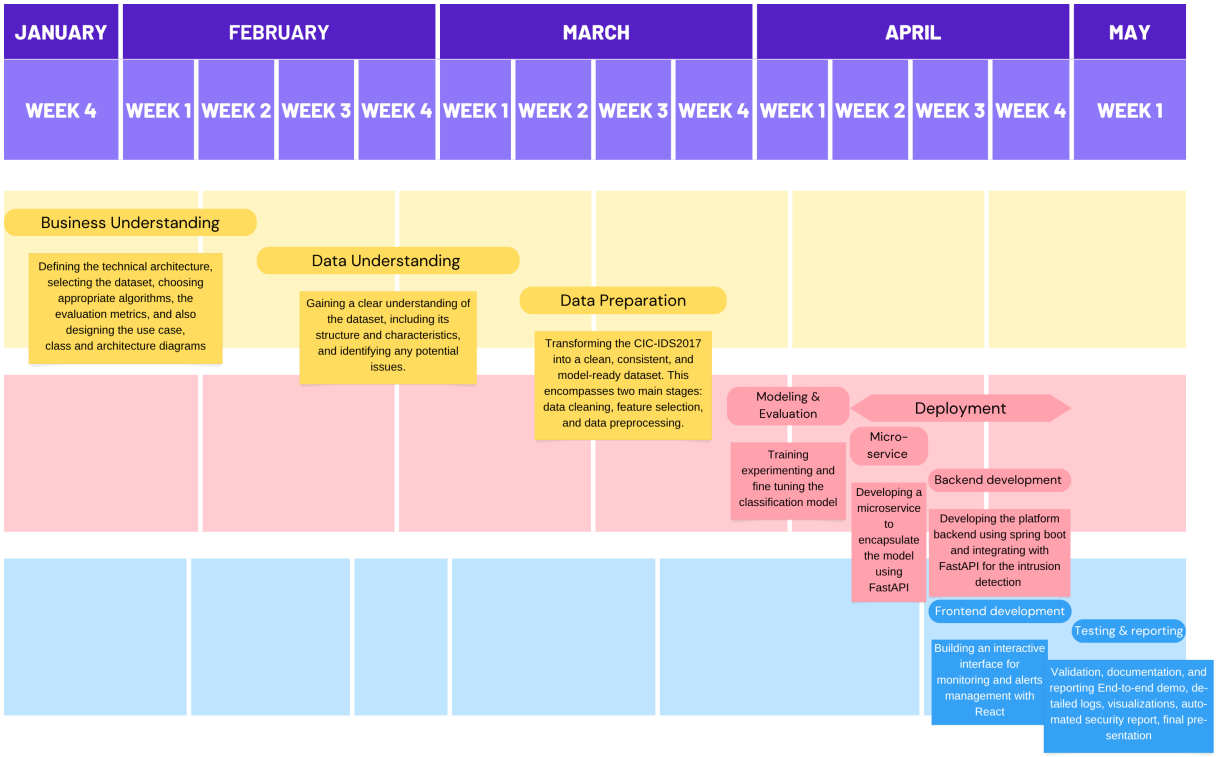


FIGURE 1.1 – Project Pipeline

1.7 Methodologies

In this section, we present the methodologies that we use in this project, including the machine learning methodology **CRISP-DM**[2] and the software development methodology **UML**[3].

1.7.1 CRISP-DM

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology is a widely used framework for structuring machine learning projects. It consists of six phases:

- **Business Understanding:** Defining project objectives and requirements and gaining a clear understanding of the business problem.
- **Data Understanding:** Collecting and exploring the data to gain insights and assess its quality by identifying data quality issues.
- **Data Preparation:** Cleaning and preparing the data for modeling by selecting relevant features, handling missing values and splitting the dataset.
- **Modeling:** Selecting and applying appropriate modeling techniques, building, testing and tuning models to optimize their performance.

- **Evaluation:** Evaluating the model's performance and determining if it meets the business objectives.

- **Deployment:** Deploying the model into production, monitoring and maintaining the model's performance.

Figure 1.2 illustrates these phases and their relationships.

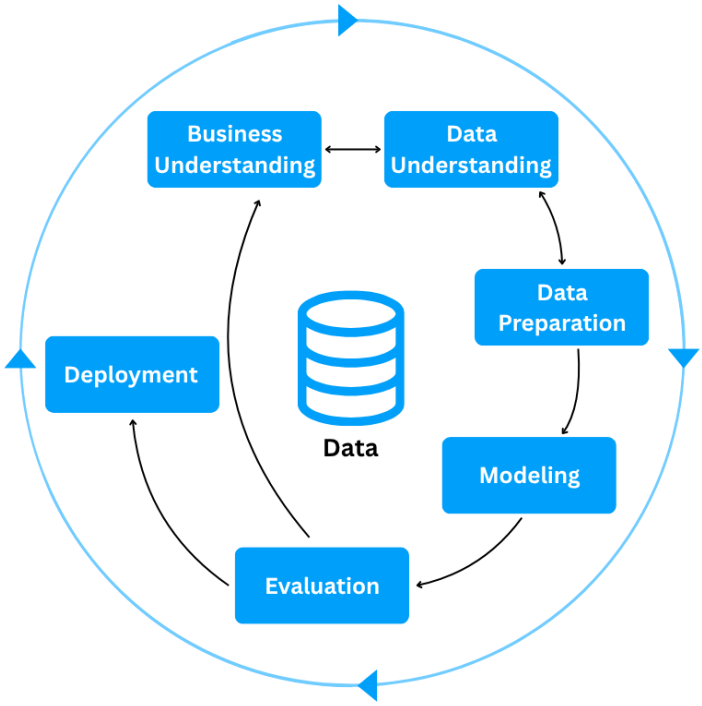


FIGURE 1.2 – CRISP-DM Methodology

1.7.2 UML

For the modeling of the software system, we use UML (Unified Modeling Language), a standardized modeling language used to specify, visualize, construct, and document the artifacts of software systems through diagrams such as use case, class, and sequence diagrams. There are two types of UML diagrams:

- **Behavioral Diagrams:** State, activity, use case, sequence, communication and interaction overview diagrams.

- **Structural Diagrams:** Class, object, component, deployment and package diagrams.

UML makes complex systems easier to design and communicate, it provides a common language for developers, architects, and stakeholders.

1.8 Conclusion

In this chapter, we introduce the host company ITXTREE, study existing network intrusion detection systems, discuss their shortcomings and finally present a solution. In the next chapter we specify the system requirements.

Chapter 2

Requirements Specification

2.1 Introduction

Throughout this chapter, we specify both the functional and non-functional requirements, identify the system actors, and present the use case diagram. By defining these elements, we aim to establish a clear and shared understanding of the system's expected behavior and constraints, providing a solid foundation for the subsequent design and implementation phases.

2.2 Functional Requirements

Functional requirements define *what* the system should perform. The main functionalities of our system are as follows:

— Authenticate;

— View Dashboard;

— View Statistics summary;

— Trigger Network Analysis;

— View Alerts List;

— View Reports;

2.3 Non-Functional Requirements

Non-functional requirements define *how* the system should perform its functions. They describe the quality attributes and constraints of the system:

- **Performance:** The system must process network traffic and generate analysis results with minimal latency, ensuring fast detection of potential intrusions.

- **Security:** The system must ensure secure authentication and enforce role-based access control, preventing unauthorized actions and protecting system integrity, while supporting non-repudiation to guarantee accountability and traceability of all operations.

- **Maintainability:** The system should be easy to maintain, allowing for efficient debugging, updates, and modifications to existing components.

- **Usability:** The system should provide an intuitive interface, enabling users to efficiently navigate and interact with its functionalities.

2.4 Actors Identification

Actors are external entities, users, organizations, or external systems that interact with the system to achieve one or more functional requirements, with each actor representing a specific role.

In the context of our project, we identify two primary actors, as illustrated in Figure 2.1

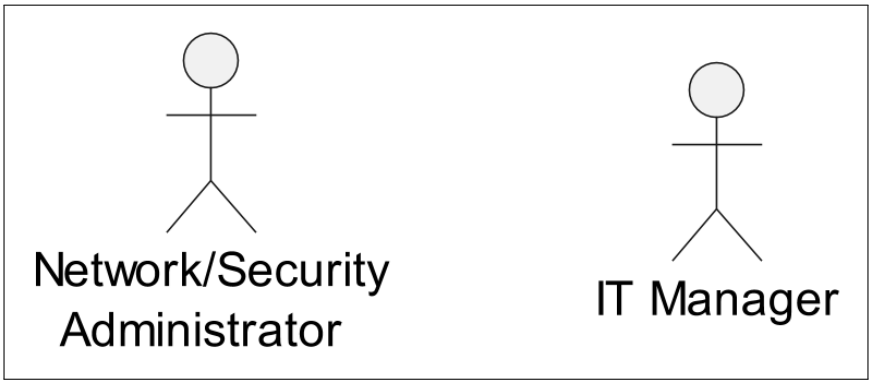


Figure 2.1 – Actors

Table 2.1 presents the roles of each actor in our system.

Table 2.1 – Roles of Actors

Actor	Role
Network/Security Administrator	- Authenticate - View Dashboard - View Statistics Summary - Trigger Network Analysis - View Alerts List
IT Manager	- Authenticate - View Dashboard - View Statistics Summary - View Reports

2.5 Use Case Diagram

A use case diagram is a visual representation of *how* users interact with the system and *what* a system does through different use cases. Figure 2.2 illustrates that diagram.

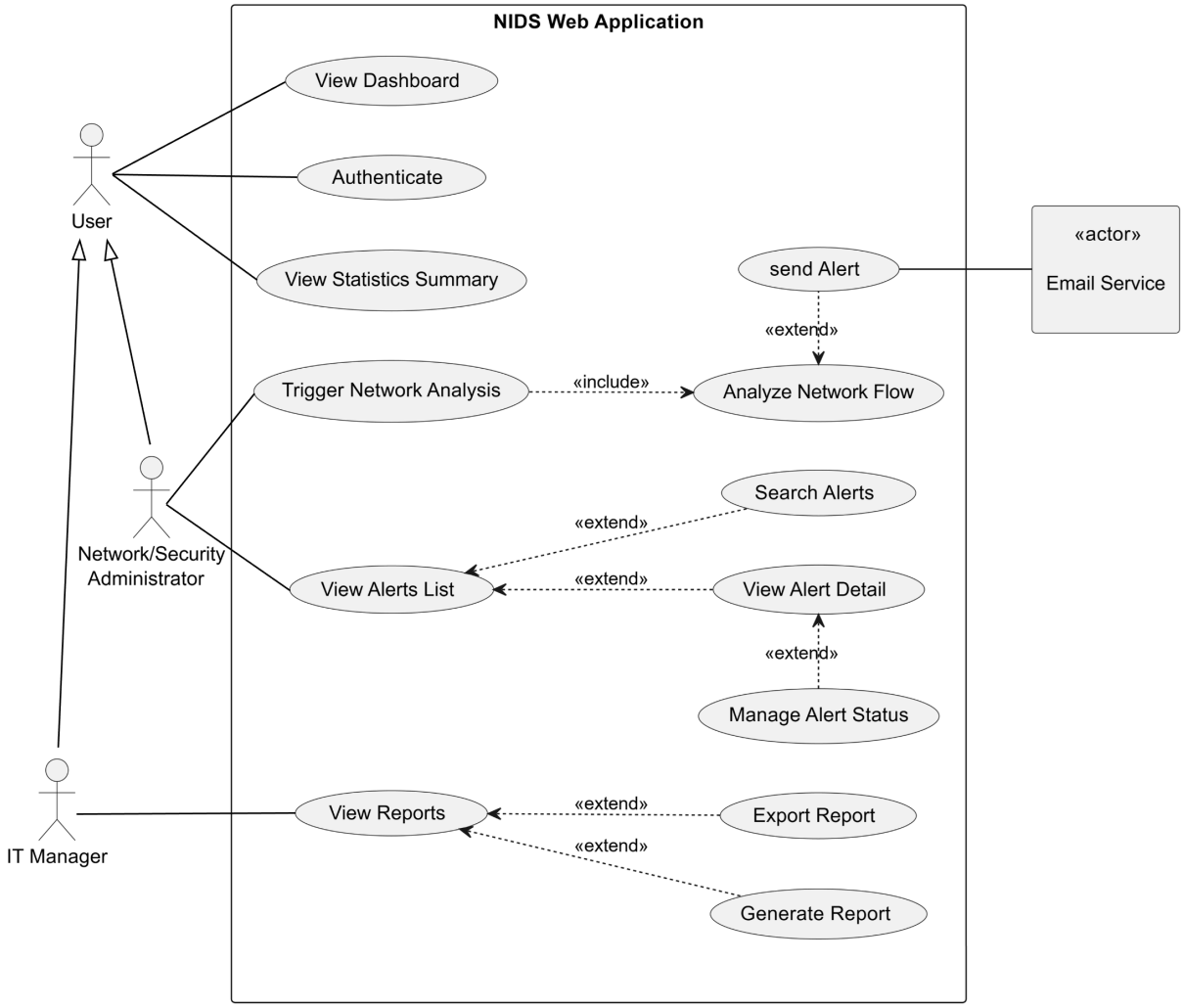


FIGURE 2.2 – Use Case Diagram

2.6 Conclusion

In this chapter, we outline the functional and non-functional requirements of our system, identify the primary actors, and represent the way they interact with the system and the system's functionality via a use case diagram. In the next chapter, we present the conceptual design of the platform.

Chapter 3

Conceptual Design

3.1 Introduction

This chapter focuses on the conceptual design of the application. It presents the sequence diagrams, the class diagram, and the deployment diagram. Its objective is to provide a clear representation of the system structure and the interactions between its different components.

3.2 Sequence Diagrams

Sequence diagrams visualize the chronological order of interactions between system components over time. Figures 3.1 and 3.2 present two key interaction scenarios of our application.

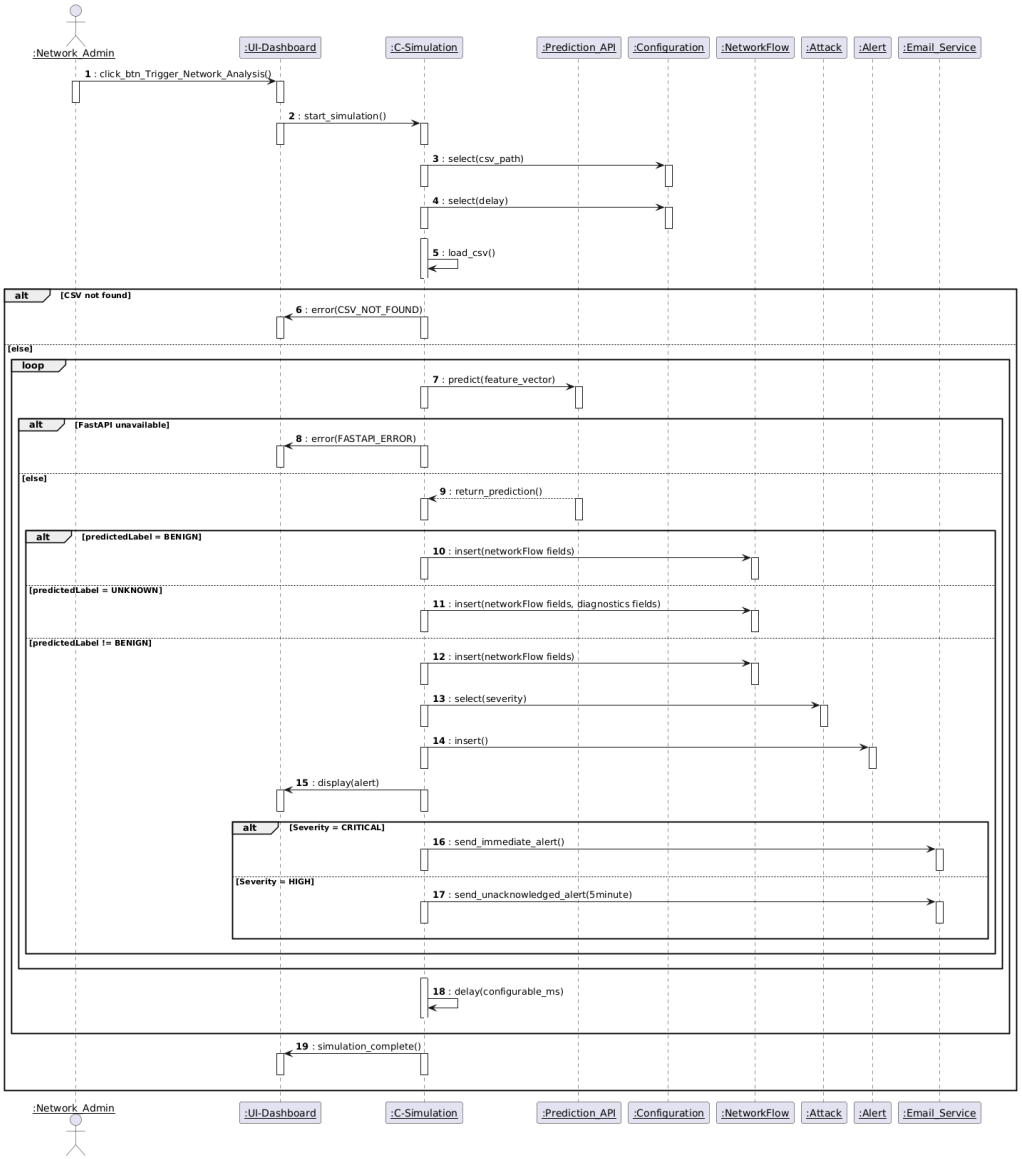


FIGURE 3.1 – Sequence diagram of the use case *Trigger Network Analysis*

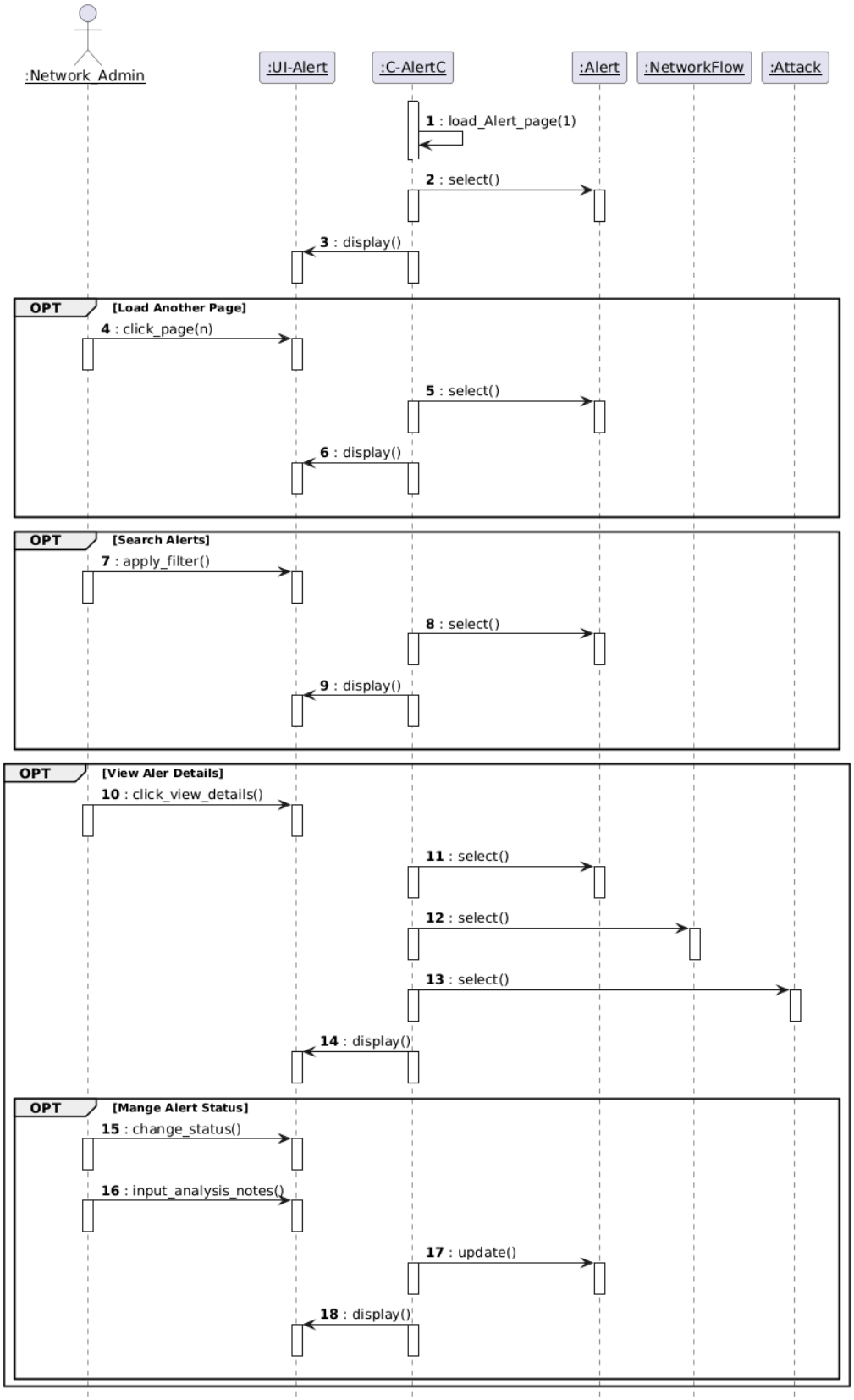


FIGURE 3.2 – Sequence diagram of the use case *View Alerts List*

3.3 Class Diagram

A class diagram describes the structure of a system by showing the system's classes, their attributes and the relationships among the objects[3]. The Figure 3.3 illustrates the class diagram of our project.

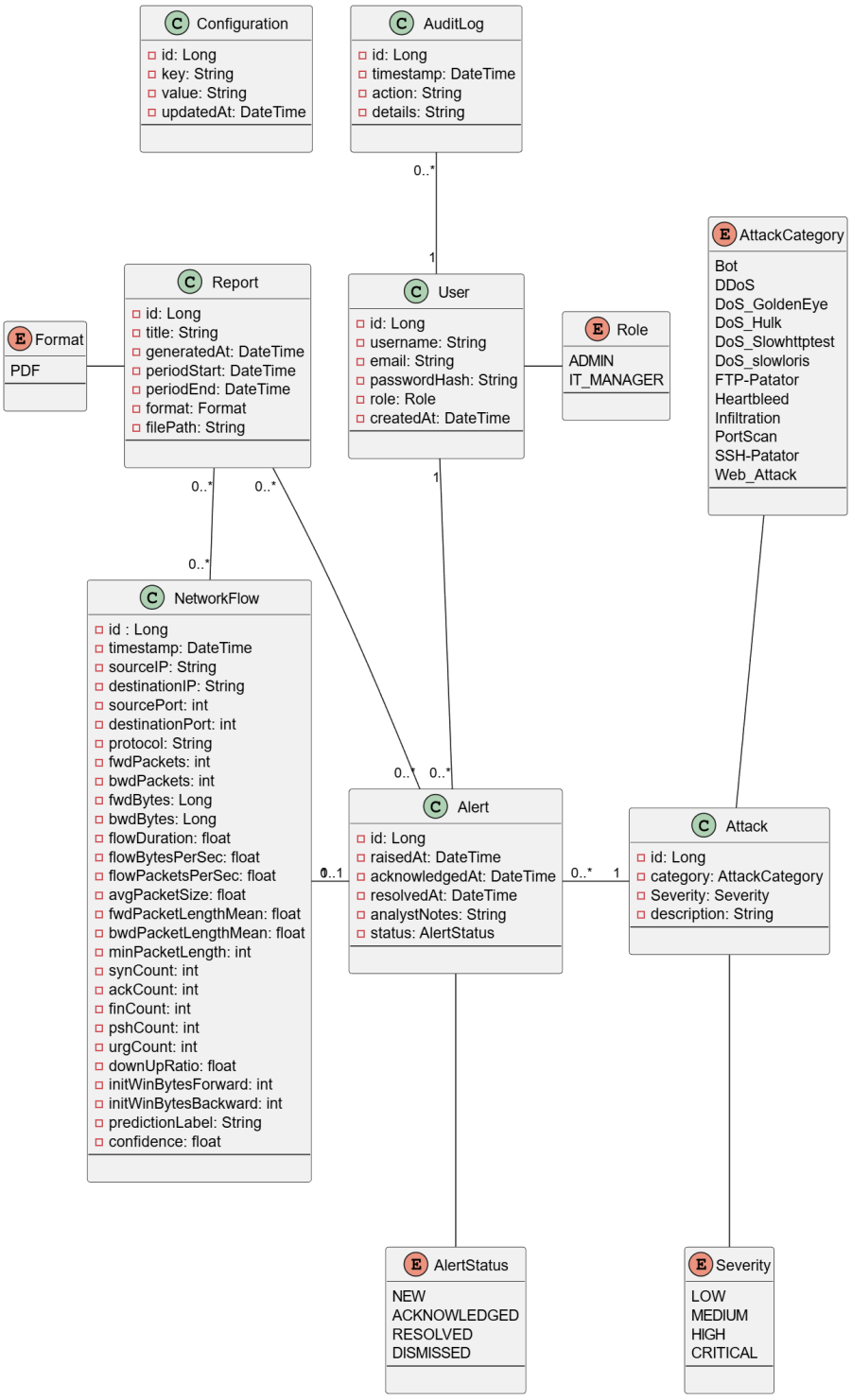


FIGURE 3.3 – Class Diagram

3.4 Conclusion

In this chapter, we concluded the conceptual design phase using UML diagrams to model the system. The following chapter marks the first phases of the The CRISP-DM methodology, the data

Chapter 4

Data Collection and Preparation

4.1 Introduction

Data constitutes the foundation of any AI system; the quality of all subsequent stages, including model training and performance, is directly dependent on it. For this reason, it requires particular attention.

This chapter presents the complete data workflow adopted in this project. It begins with data collection, followed by an in-depth exploration phase aimed at understanding the structure, characteristics, and underlying patterns of the dataset. Based on the insights obtained during this stage, we then proceed to data preprocessing, where appropriate transformations and refinements are applied to ensure the dataset is properly structured and suitable for model training.

4.2 Data Collection

This section introduces the process of acquiring the data used in this project. In general, this phase may involve either constructing a dataset from scratch or relying on existing sources. In our case, the decision to use a public dataset is mainly dictated by the nature of the cybersecurity domain.

Unlike other fields, network intrusion data cannot be collected on demand, as it depends on the occurrence of unpredictable and potentially harmful attacks. In addition, data collection in real operational environments is often constrained due to privacy concerns, security risks, and limited access to organizational traffic. Furthermore, even when such data is available, obtaining reliable ground truth labels is difficult and typically requires extensive manual analysis.

For these reasons, constructing a proprietary dataset is impractical, particularly in the absence of real network traffic provided by the host organization. Consequently, the use of a public dataset becomes a necessary and coherent choice, as such datasets are generated in controlled environments and include labeled instances based on expert-defined attack scenarios.

4.2.1 Dataset comparative

We narrowed it down to these three datasets. Below is a comparative table 4.1 using our primary selection criteria.

Table 4.1 – Comparison of network intrusion detection datasets

Dataset	Source	Size	Data Generation Approach	Attack Diversity and Temporal Relevance	Usability
NSL-KDD [4]	Canadian Institute for Cybersecurity at the University of New Brunswick (UNB)	Small (≈148k records)	Derived from the KDD '99 dataset, its traffic was generated in a military-style network environment with predefined automated attack scripts.	Low; 4 attack families: DoS, Probe, R2L, U2R with several specific attacks in each family.	High; legacy benchmark widely used in academic studies, mainly for baseline comparison and educational purposes.
UNSW-NB15 [5]	Australian Centre for Cyber Security (ACCS), UNSW Canberra	Medium (≈2.5M records)	Synthetic traffic generated with IXIA PerfectStorm tool.	Medium; 9 attack types: Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms.	High; widely used modern benchmark for machine learning-based intrusion detection research.
CIC-IDS-2017 [6]	Canadian Institute for Cybersecurity at the University of New Brunswick (UNB)	Medium (≈2.8M records)	Traffic generated in an enterprise-like network environment composed of attacker and victim machines, using real attack tools.	High; 14 attack types: Includes the most common attacks based on the 2016 McAfee report, such as Web based, Brute force, DoS, DDoS, Infiltration, Heart-bleed, Bot and Scan.	Very High; one of the most widely adopted and current standard datasets in intrusion detection research.

The comparative analysis of the three datasets reveal that each has its own limitations. NSL-KDD, although still relevant, exhibits limitations in size and temporal relevance, as it is relatively old and does not reflect recent network behavior and modern attacks types. UNSW-NB15 is larger and offers a more recent alternative, but it remains limited in terms

of attack type coverage, and its traffic is fully synthetically generated. This makes CIC-IDS2017 the most suitable dataset, as it provides a more realistic experimental setting, a wider range of the most common attack types, and it is also widely adopted in recent research studies.

4.2.2 Dataset overview

CIC-IDS2017 was created by the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick (UNB), within a controlled environment, where the network traffic was captured in packet capture (PCAP) format.

The transformation of raw PCAP files was done using a feature extraction tool, CICFlowMeter. Therefore a row in this dataset doesn't represent a packet, it represents an overview of the session behaviour between two endpoints through statistical summary.

4.3 Data Understanding

The Data Understanding phase of the CRISP-DM methodology focuses on loading, describing, and analyzing the dataset to build a solid foundation for the rest of the project. This section dives into the four tasks of this phase: `Initial Exploration`, `Data Description`, `Data Exploration`, and `Data Quality`, with each task targeting an aspect of the data, from its structure and content to its quality.

4.3.1 Initial Exploration

The CIC-IDS2017 dataset is captured over five days, it is distributed across 8 files split by day and session: `Monday`, `Tuesday`, and `Wednesday` each corresponding to a single file, while `Thursday` and `Friday` are split into AM and PM sessions. The `Friday PM` session is further divided into two files to isolate the two attacks, `DDoS` and `PortScan`.

4.3.2 Data Description

As mentioned in the previous subsection, the dataset is split into 8 files as shown in Table 4.2.

Table 4.2 – Dataset Files

File	Number of Rows	Number of Columns
Monday	529,918	79
Tuesday	445,909	79
Wednesday	692,703	79
Thursday AM	170,366	79
Thursday PM	288,602	79
Friday AM	191,033	79
Friday PM (DDoS)	225,745	79
Friday PM (PortScan)	286,467	79
Total	2,830,743	—

The dataset contains a total of 2,830,743 rows across the 8 files. All files have consistent columns, which is expected as they are generated by the same CICFlowMeter pipeline. This consistency allows for seamless concatenation of the files into a single DataFrame for the subsequent analysis.

- **Whitespace Issue:** Leading white spaces is found in columns names and are stripped across all files as they can raise `KeyError` exceptions when accessing columns by name.

- **Corrupted Labels:** Corrupted labels are discovered in the Thursday AM file, where three labels contain mojibake character instead of an em-dash (-) caused by an encoding issue as shown in Figure 4.1 and are therefore corrected.

```
[Thursday-AM]  'Web Attack ◆ Brute Force'
[Thursday-AM]  'Web Attack ◆ XSS'
[Thursday-AM]  'Web Attack ◆ Sql Injection'
```

FIGURE 4.1 – Label Corruption

- **Feature Description:** The dataset contains 79 columns, of them 78 are numeric level-flow features and the last one is the target Label. These features can be grouped into 12 categories as shown in Table 4.3.

TABLE 4.3 – Feature Groups

Group	Features	Description
Flow Metadata	`Destination Port`, `Flow Duration`	Total duration of the flow and the destination port
Packet/Byte Count	`Total Fwd Packets`, `Total Backward Packets`, `Total Length of Fwd Packets`, `Total Length of Bwd Packets`, `act_data_pkt_fwd`	Number of packets and packet bytes in both forward (from source to destination) and backward directions (from destination to source), and packets having actual data in the forward direction.
Packet/Segment Length Statistics	`Fwd Packet Length Max/Min/Mean/Std`, `Bwd Packet Length Max/Min/Mean/Std`, `Max/Min Packet Length`, `Packet Length Mean/Std/Variance`, `Average Packet Size`, `Avg Fwd Segment Size`, `Avg Bwd Segment Size`	Packets and segments length statistics (e.g. max, min, mean, std etc...).
Flow Rate	`Flow Bytes/s`, `Flow Packets/s`, `Fwd Packets/s`, `Bwd Packets/s`, `Down/Up Ratio`	Flow throughput, forward and backward packet rates, flow asymmetry ratio (backward packets (download) to forward packets (upload)).

Inter-Arrival Time	`Flow IAT Max/Min/Mean/Std`, `Fwd IAT Max/Min/Mean/Std/Total`, `Bwd IAT Max/Min/Mean/Std/Total`	Time between consecutive packets in the flow, forward, and backward directions.
TCP Flag Count	`FIN Flag Count`, `SYN Flag Count`, `RST Flag Count`, `PSH Flag Count`, `ACK Flag Count`, `URG Flag Count`, `CWE Flag Count`, `ECE Flag Count`, `Fwd PSH Flags`, `Bwd PSH Flags`, `Fwd URG Flags`, `Bwd URG Flags`	TCP flags counts for the entire flow, forward and backward directions.
Header Length	`Fwd Header Length`, `Bwd Header Length`, `Fwd Header Length.1`	Backward and forward header lengths.
Bulk Transfer Statistics	`Fwd Avg Bytes/Bulk`, `Fwd Avg Packets/Bulk`, `Fwd Avg Bulk Rate`, `Bwd Avg Bytes/Bulk`, `Bwd Avg Packets/Bulk`, `Bwd Avg Bulk Rate`	Average forward and backward bulk (a sequence of consecutive packets in the same direction) transfer statistics.
Subflow Packet/Byte Count	`Subflow Fwd Packets`, `Subflow Fwd Bytes`, `Subflow Bwd Packets`, `Subflow Bwd Bytes`	Number of packets and packet bytes in both directions in a subflow.
Window/Segment Size	`Init_Win_bytes_forward`, `Init_Win_bytes_backward`, `min_seg_size_forward`	Window sizes in both the forward and backward directions and the minimal segment size in the forward direction.
Active/Idle Time	`Active Max/Min/Mean/Std`, `Idle Max/Min/Mean/Std`	Duration of active and idle periods within a flow.
Target	`Label`	The target label indicating the class the flow belongs to.

- **Duplicate Column:** The column `Fwd Header Length.1` is identified as a duplicate of `Fwd Header Length`, confirmed by the fact that both contain the same values.

- `Packet Length Mean` and `Average Packet Size` both measure the average packet size, yet the two are equal only when zero. This inequality is caused by a bug in the denominator of `Packet Length Mean`, as it uses the internal `flowLengthStats` accumulator to get its value and the first packet is added twice to it, inflating its packet count by one, while `Average Packet Size` divides by the correctly computed total packet count [4].

All 54 features are of type `int64`, 24 features are of type `float64`, with the last feature, the target *Label*, being of type `object`.

Table 4.4 presents the statistics of some of the features.

Table 4.4 – Feature Statistics

Feature	Min	Max	Mean	Std
Flow Duration	-13	119,999,998	14,785,663.92	33,653,744.09
Total Backward Packets	0	121,922	10.39	997.39
act_data_pkt fwd	0	213,557	5.42	636.43
Min Packet Length	0	1,448	16.43	25.24
Flow Bytes/s	-241,000,000	inf	inf	nan
Flow IAT Max	-13	120,000,000	9,182,475.32	24,459,539.25
Bwd PSH Flags	0	0	0	0
Fwd Header Length	-322,212,234,632	4,644,908	-25,997.39	21,052,857.87
Bwd Avg Bulk Rate	0	0	0	0

Some of the feature has negative, inf and constant statistic of zero. Those features are further investigated in the data quality section.

- **Class Distribution:** The dataset contains 15 classes, BENIGN and 14 attacks, Table 4.5 shows the distribution of these classes across the dataset files.

Table 4.5 – Class Distribution Per File

File	Class	Class Count
Monday	BENIGN	529,918
Tuesday	BENIGN	432,074
	FTP-Patator	7,938
	SSH-Patator	5,897
Wednesday	BENIGN	440,031
	DoS Hulk	232,443
	DoS GoldenEye	10,293
	DoS Slowloris	5,796
	DoS Slowhttptest	5,499
	Heartbleed	11
Thursday AM	BENIGN	168,186
	Web Attack - Brute Force	1,507
	Web Attack - XSS	652
	Web Attack - Sql Injection	21
Thursday PM	BENIGN	288,566
	Infiltration	36
Friday AM	BENIGN	189,067
	Bot	1,966
Friday PM (DDoS)	DDoS	128,027
	BENIGN	97,718
Friday PM (PortScan)	PortScan	158,930
	BENIGN	127,537

With the exception of Monday, which contains only BENIGN traffic, all files contain at least one attack class alongside BENIGN samples. Each attack class is present in only one file.

Table 4.6 provides a description of the 14 attack classes of the dataset.

Table 4.6 – Attack Classes

Class	Description
FTP-Patator (File Transfer Protocol Patator)	A brute force attack using the Patator tool targeting FTP servers, an old unencrypted protocol that transmits data and credentials in plain text, using trial and error to guess credentials in order to gain unauthorized access.
SSH-Patator (Secure Shell Patator)	A brute force attack using the Patator tool targeting SSH servers, a secure encrypted protocol that encrypts the entire session including the login, using trial and error to gain unauthorized access to individual accounts.
DoS Hulk	A Denial-of-Service attack using the HULK tool, which is used to overwhelm web servers by generating a large number of unique randomized requests at irregular intervals.
DoS GoldenEye	A Denial-of-Service attack using the GoldenEye tool, which dynamically generates multiple parallel connections (multi-threaded) to a target URL while keeping them alive to exhaust the available connection slots, simultaneously forcing the server to bypass caching mechanisms and fully process every request.
DoS slowloris	A Denial-of-Service attack using the Slowloris tool, which enables a single machine to overwhelm a web server by opening multiple connections and keeping them open by sending partial requests (only headers) periodically, while requiring very low bandwidth, hence its name, as it operates in a low and slow manner.
DoS Slowhttptest	A Denial-of-Service attack using the SlowHTTPTest tool, which creates very slow connections by sending seemingly legitimate requests using low bandwidth and intentionally delaying their completion, thereby consuming the limited number of concurrent connections the server can handle.
DDoS	An attack used to overwhelm the target system with a flood of traffic from multiple sources, making it unable to respond to legitimate requests, potentially leading to a complete shutdown of the system.

Bot	An attack carried out using the ARES botnet, a malware-based network of compromised computers built on a RAT (Remote Access Trojan) and botnet framework, which allows the attacker to take control of infected machines and coordinate them for malicious purposes. It first infects victim machines using phishing emails or malicious downloads, after which the infected machines connect back to the attacker's Command and Control (C&C) server. Finally, the attacker issues commands to all bots, that is, all infected machines, simultaneously.
Heartbleed	An attack in which the attacker exploits the Heartbleed bug in the OpenSSL cryptographic library to read sensitive memory contents by sending malformed heartbeat requests demanding more memory than the actual payload from the server running the vulnerable version of OpenSSL, after which the server allocates the requested memory without any verification and returns it, exposing whatever sensitive data happened to be stored in that memory chunk at that moment. It is used to steal private encryption keys, usernames and passwords, and to intercept session tokens, permitting session hijacking by providing the stolen token to the server so that the attacker is treated as a legitimate logged-in user, as well as to decrypt previously captured encrypted traffic that the attacker had intercepted beforehand.
Infiltration	A multi-stage attack to gain unauthorized access to systems, in which the attacker first deploys a malicious file disguised as a legitimate one, such as a document or a downloadable link, containing a backdoor. The victim executes the file unknowingly, after which the backdoor establishes a reverse connection back to the attacker's machine, granting the attacker remote control over the compromised system. Once inside, the attacker scans the victim's internal network using tools (e.g. Nmap) to discover live hosts (any active and reachable device on the network), as well as open ports and running services. This allows the attacker to map the internal network and identify potential targets for further infiltration. The attacker then moves laterally by infiltrating another machine, spreading the attack further across the network. Finally, an attempt is made to exfiltrate sensitive data (transferring confidential information out of the network), and to escalate privileges on the compromised systems in order to gain access to even more restricted and sensitive resources.

PortScan	An attack used to discover open ports and whether they are receiving or sending data, in addition to potential vulnerabilities that could be exploited.
Web Attack - Brute Force	An attack that uses a forceful approach to steal credentials to gain unauthorized access to systems. It can be carried out in many ways, from trying common credential combinations manually, to using dictionary attacks, or a combination of both.
Web Attack - XSS	An attack in which malicious scripts are injected into trusted websites, by exploiting their input validation vulnerability, and executed by other end users' browsers, which have no way to distinguish it from legitimate content.
Web Attack - Sql Injection	An attack in which the attacker interferes with the queries that an application makes to its database enabling him to view data that he is not supposed to have access to, including data belonging to other users and any other data that the platform has access to, as well as modify or delete it. It is carried out by injecting malicious SQL queries into an input field, which the vulnerable application passes directly to the database as a query instead of treating it as plain data. The database then executes it, allowing the attacker to manipulate the database in unintended ways.

4.3.3 Data Exploration

Table 4.7 provides a comprehensive overview of the classes' distribution across the dataset.

TABLE 4.7 – Class Distribution

Class	file	Count	Percentage
BENIGN	Monday	529,918	18.72%
	Tuesday	432,074	15.26%
	Wednesday	440,031	15.54%
	Thursday AM	168,186	5.94%
	Thursday PM	288,566	10.19%
	Friday AM	189,067	6.68%
	Friday PM (DDoS)	97,718	3.45%
	Friday PM (PortScan)	127,537	4.51%
	Total	**2,273,097**	**80.3%**
DoS Hulk	Wednesday	231,073	8.163%
PortScan	Friday PM (PortScan)	158,930	5.614%
DDoS	Friday PM (DDoS)	128,027	4.523%
DoS GoldenEye	Wednesday	10,293	0.364%

FTP-Patator	Tuesday	7,938	0.280%
SSH-Patator	Tuesday	5,897	0.208%
DoS slowloris	Wednesday	5,796	0.205%
DoS Slowhttptest	Wednesday	5,499	0.194%
Bot	Friday AM	1,966	0.069%
Web Attack - Brute Force	Thursday AM	1,507	0.053%
Web Attack - XSS	Thursday AM	652	0.023%
Infiltration	Thursday PM	36	0.0013%
Web Attack - Sql Injection	Thursday AM	21	0.0007%
Heartbleed	Wednesday	11	0.0004%
Total		**2,830,743**	**100%**

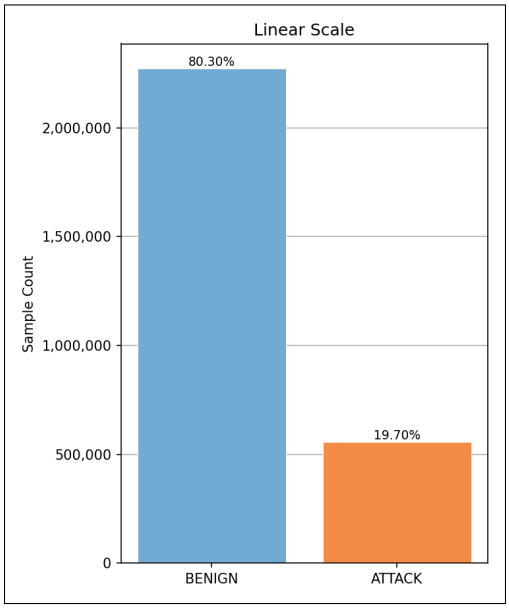


FIGURE 4.2 – BENIGN vs Attack Ratio

Figure 4.2 visualizes the BENIGN vs Attack ratio. The BENIGN class is the dominant class in the dataset, accounting for 80.3%, while the remaining 15 attack classes collectively amount to only 19.7%.

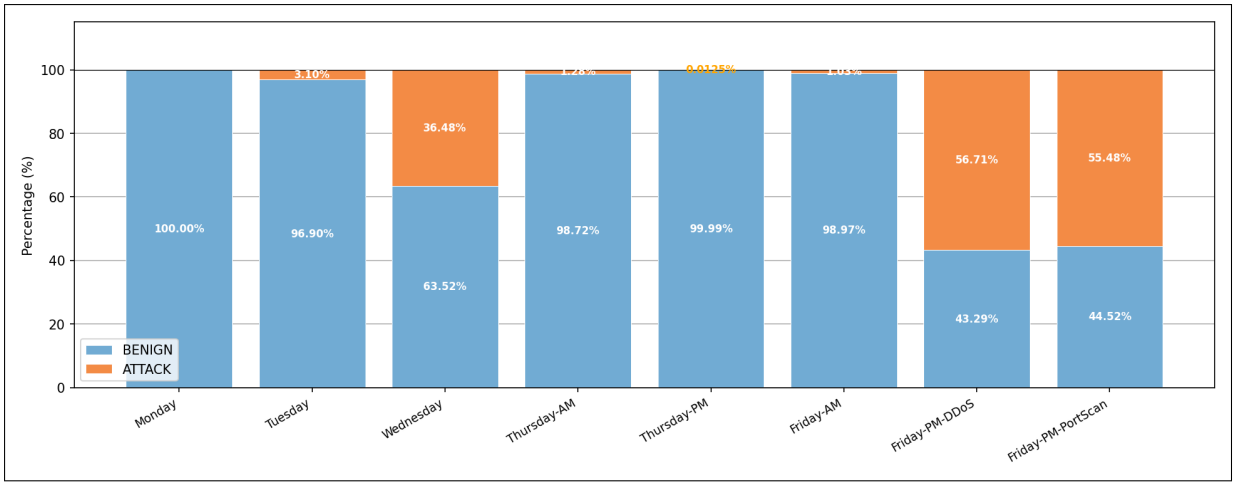


FIGURE 4.3 – Benign vs Attack Ratio per file

Figure 4.3 shows the BENIGN vs Attack ratio per file. The BENIGN class dominates across all files, with the only exceptions being the two Friday PM files, where attack traffic exceeds BENIGN, with attack ratios of 56.7% and 55.5% in the Friday PM DDoS and Friday PM PortScan respectively. The BENIGN ratio ranges from 100% in the Monday file down to 43.3% in the Friday PM DDoS file. Several files show very low attack ratios, such as Thursday AM at 1.3% and Friday AM at 1%, while the Thursday PM file has an attack ratio so low at 0.012% that it is invisible in the figure. Figure 4.4 visualizes the

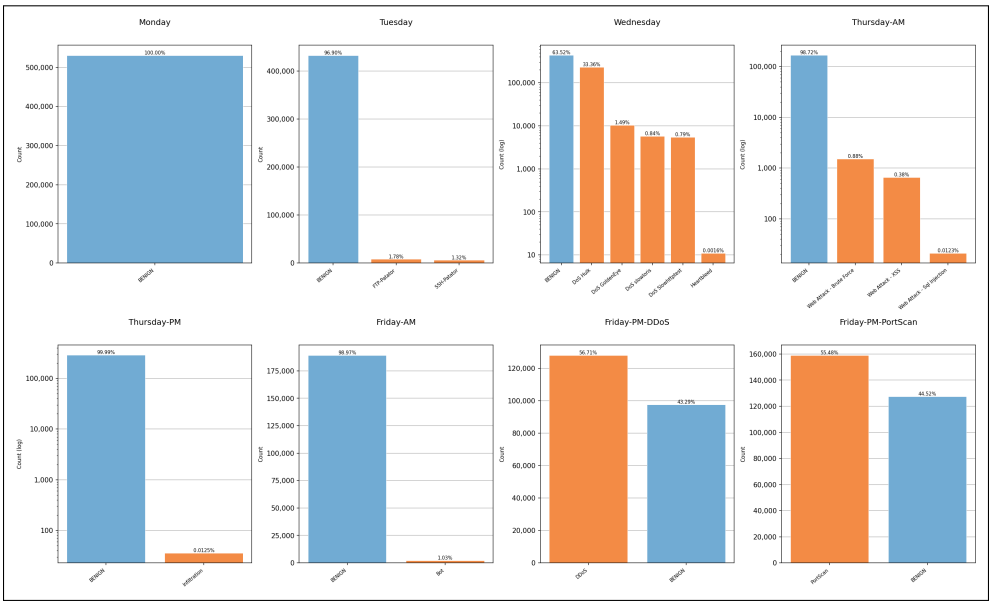


FIGURE 4.4 – Class distribution per file

class distribution per file. As mentioned previously, each attack class is present in only one file, with each file containing between 0 and 5 attack classes, most files containing only one. The two brute force attacks, FTP-Patator and SSH-Patator are both present in the Tuesday file. The DoS attacks and Heartbleed are grouped in the Wednesday file, and the web attacks are present in the Thursday AM file. The remaining classes each appear exclusively in one of the other files.

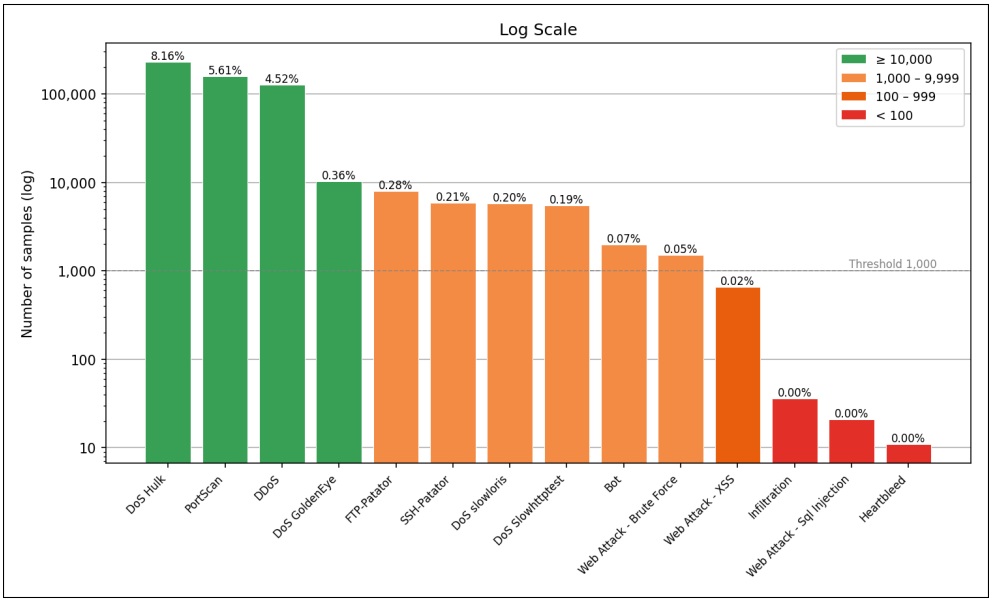


FIGURE 4.5 – Attack distribution

Figure 4.5 visualizes the global attack distribution. It varies significantly, ranging from DoS Hulk at 8.16% down to Heartbleed at 0.0004%, with the three rarest classes, Infiltration, Web Attack - Sql Injection, and Heartbleed, collectively accounting for only 0.0024%.

This distribution is intentional, as the dataset was generated to imitate real-world network traffic, where the vast majority of traffic is normal and only a small fraction are attacks. However, this introduces a severe class imbalance that can bias the model towards the majority class, leading it to treat BENIGN as the default outcome. As a result, the model may perform poorly on the attack classes and potentially fail to learn their distinguishing patterns altogether.

The class imbalance extends beyond the BENIGN class and is also present among the attack classes themselves. DoS Hulk, PortScan, DDoS and DoS GoldenEye each exceed 10,000 samples, providing the model with sufficient examples to reliably learn their patterns. FTP-Patator, SSH-Patator, DoS slowloris, DoS Slowhttptest, Bot and Web Attack - Brute Force fall in a moderate range, each with over 1,000 samples, which while on the lower end, should still allow the model to capture their general patterns. Web Attack - XSS class, with only 652 samples sits in a more concerning range where learning becomes unreliable.

The three remaining classes represent the most critical imbalance: Infiltration with 36 samples, Web Attack - Sql Injection with 21 samples, and Heartbleed with only 11 samples. At this scale, the model is unlikely to learn their patterns meaningfully, and even in cases where it does, generalization to unseen data is very unlikely, leading to poor performance on new data.

After exploring the class distribution, we turn to the feature space by computing the correlation matrix for the 78 features of the dataset. It is a square matrix where each element represents the linear association between two features, with values ranging from -1 to 1. Positive values indicate that both features increase together, while negative values indicate that one increases as the other decreases. In both cases the strength of the correlation increases as the value approaches 1 or -1, while a value of zero indicates no correlation. The diagonal is always 1 as every feature is perfectly correlated with itself.

The correlation matrix is computed using the Pearson Correlation Coefficient (PCC), also known as Pearson's r:

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{4.1}$$

where x and y are the two features being compared, $\bar{x}$ and $\bar{y}$ are their respective means, and n is the number of samples.

The absolute value of the correlation matrix was taken to focus on the strength of the correlation regardless of its direction. A threshold of 0.95 was used to identify highly correlated feature pairs, where any pair with a correlation coefficient of 0.95 or higher is considered redundant, meaning both features essentially carry the same information, providing no additional information to the model.

39 highly correlated pairs are identified Figure 4.6 visualizes their correlation matrix.

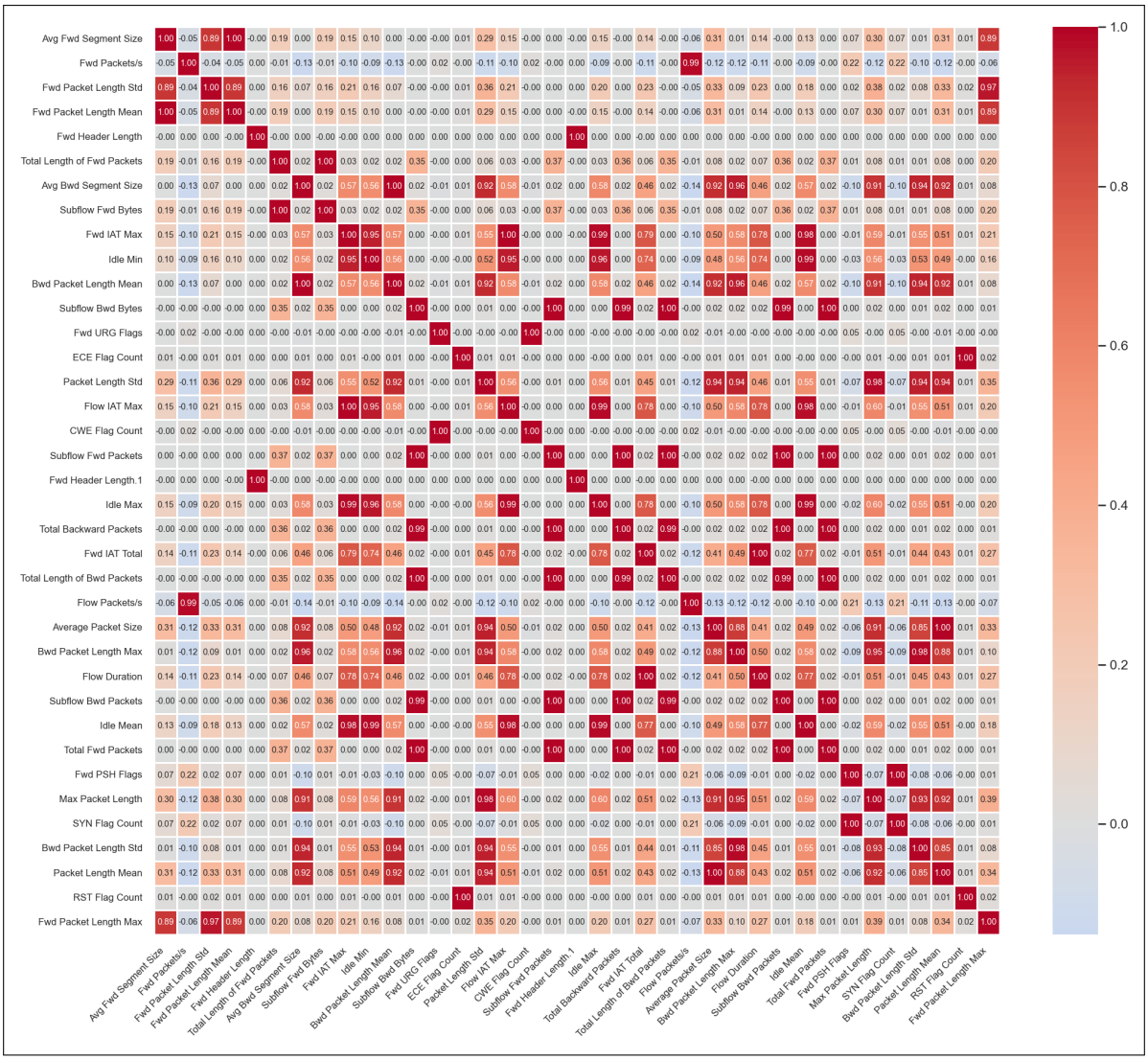


FIGURE 4.6 – Correlation Heatmap

8 pairs have a perfect correlation of 1, meaning they are perfectly linearly correlated. Table 4.8 lists these perfectly correlated pairs.

TABLE 4.8 – Perfectly Correlated Feature pairs

No.	Feature 1	Feature 2
1	Bwd Packet Length Mean	Avg Bwd Segment Size
2	Fwd Packet Length Mean	Avg Fwd Segment Size
3	Total Fwd Packets	Subflow Fwd Packets
4	Total Backward Packets	Subflow Bwd Packets
5	Total Length of Bwd Packets	Subflow Bwd Bytes
6	Fwd PSH Flags	SYN Flag Count
7	Fwd URG Flags	CWE Flag Count
8	Fwd Header Length	Fwd Header Length.1

- **First and second pairs:** `Bwd Packet Length Mean` and `Avg Bwd Segment Size`, `Fwd Packet Length Mean` and `Avg Fwd Segment Size` going by name the two cal-

culates the average size of two different layers of the OSI model, the former belonging to the network layer it is composed of Network Header (IP header) + Segment, while the latter belongs to the transport layer and is composed of Transport Header (TCP/UDP Header) + Payload, but in the CICFlowMeter they are both calculated as the average size (bytes) of the payload in the backward/forward direction, which means there is no difference between a segment and a packet making the two features a duplicate of each other.

- **Third, fourth and fifth pairs (with an additional pair):** `Total Fwd Packets` and `Subflow Fwd Packets`, `Total Backward Packets` and `Subflow Bwd Packets`, `Total Length of Fwd Packets` and `Subflow Fwd Bytes`, and `Total Length of Bwd Packets` and `Subflow Bwd Bytes` (near perfect correlation of 0.99999) nominally represent two different levels of aggregation, with the first being for the entire flow and the second being for a subflow. A subflow is detected when the inter-arrival time (IAT) between two consecutive packets exceeds 1 second, with the subflow features computed by dividing the flow totals by the number of detected subflows (`sfCount`). The assumption that all subflows contain the same number of packets or bytes is already a conceptual limitation, but the way `sfCount` is calculated is a bug in itself.

  Due to a misplaced parenthesis in the source code, the condition that is supposed to check whether the gap between two packets exceeds 1 second fires on the first inter-packet gap that does so and never again, making `sfCount` equal to 1 for virtually every flow in the dataset. As a result, all four subflow features reduce to dividing the flow total by 1, making them equal to their corresponding totals.

  The packet-count pairs hold this equality perfectly throughout the entire dataset since integer division by 1 is exact. The byte pairs behave the same way for the vast majority of flows, with rare divergence explained not by `sfCount` but by a type mismatch: `Total Length of Fwd/Bwd Packets` is output as a `double`, while `Subflow Fwd/Bwd Bytes` is output as a `long`. For flows with very large byte totals, floating-point precision loss in the `double` path produces a value that differs slightly from the exact `long`, which is the only source of divergence observed.

  All four pairs are therefore redundant, carrying no additional information beyond their corresponding flow-level totals, a consequence of a single bug.

- **Sixth pair:** `Fwd PSH Flags` and `SYN Flag Count` are nominally designed to capture different aspects of TCP behavior, with the first counting forward packets with the PSH flag set and the second counting all packets with the SYN flag set across the flow. However, they are perfectly correlated throughout the dataset, with exactly 2,699,265 flows where both equal 0 and 131,478 flows where both equal 1, a result of two compounding causes.

  The first is a bug in the generation-time version of CICFlowMeter where flags are only updated on the first packet, reducing both features to binary indicators of whether that single packet had the respective flag set. This means the SYN-ACK from the server is never counted, contradicting what the feature name implies.

  The second is a traffic capture artifact. Flows where both equal 1 belong exclusively to attacks that establish real TCP connections (`FTP-Patator`, `SSH-Patator`,

`DoS slowloris`, `DoS Slowhttptest`, `Infiltration`) and benign traffic, while flows where both equal 0 include attacks that bypass the handshake (`DDoS`, `DoS Hulk`, `DoS GoldenEye`, `PortScan`, `Bot`, Web Attacks) and mid-stream captures. In this dataset, any flow starting with a SYN packet consistently has PSH set on that same first packet, making both features identical binary indicators of whether the flow was captured from the beginning of a TCP connection, and therefore entirely redundant.

- **Seventh pair:** `Fwd URG Flags` and `CWR Flag Count` are perfectly correlated throughout the dataset, with 2,830,428 flows where both equal 0 and only 315 flows where both equal 1, all belonging exclusively to benign traffic. As with the previous pair, the flag-counting bug reduces both features to binary indicators of the first packet only. The co-occurrence of URG and CWR on the same first packet is consistent with legitimate TCP congestion control behavior, which explains their absence in attack traffic, making both features redundant with each other.

- **Eighth pair:** `Fwd Header Length` and `Fwd Header Length.1` these two features are duplicates of each other as mentioned previously in the Data Description subsection 4.3.2, which makes sense for them to be perfectly correlated.

4.3.4 Data quality

The quality of a model's output is directly tied to its input. In this section we evaluate the dataset for any quality issues that could disrupt model training such as missing values (NaN - Not a number), infinite values (inf), negative values, duplicates rows, constant and near-constant features. For each issue, the analysis investigates the root cause, assesses its global impact on the dataset, across classes and whether it endangers any minority classes.

NaN and Infinite Values: Scanning the dataset for undefined values revealed that NaN values appear exclusively in `Flow Bytes/s`, while infinite values appear in both `Flow Bytes/s` and `Flow Packets/s`. To understand the origin, the CICFlowMeter source code was inspected directly. The relevant formulas are:

$$\text{Flow Bytes/s} = \frac{\text{forwardBytes} + \text{backwardBytes}}{\text{flowDuration}/1{,}000{,}000}$$

$$\text{Flow Packets/s} = \frac{\text{packetCount (fwd + bwd)}}{\text{flowDuration}/1{,}000{,}000}$$

Both features divide by `Flow Duration`. When `Flow Duration = 0`, any nonzero numerator yields Inf, and a zero numerator yields NaN 0/0. This distinction maps directly onto two observable row types in the dataset:

- Rows with 2 forward packets, 0 backward, and nonzero byte count:
  `Flow Duration` $= 0$ with bytes > 0, producing `Flow Bytes/s` $=$ Inf.

- Rows with 1 forward packet, 1 backward packet, and zero byte count:
  `Flow Duration` $= 0$ with zero bytes, producing `Flow Packets/s` $=$ NaN.

The zero byte count in the second case, despite existing packets, is because CICFlowMeter computes `Total Length of Fwd/Bwd Packets` from the packet payload only. Packets carrying no data content only TCP headers (ACK, SYN) produce zero payload length. The true anomaly in both cases is `Flow Duration = 0`.

`Flow Duration` is computed as the timestamp difference between the last and first packet in a flow. It evaluates to zero when two packets arrive within the same timestamp resolution window as the tool lacks the granularity to distinguish arrivals that are nanoseconds apart. Every affected row contains exactly two packets total, either two forward or one forward and one backward, which is consistent with this explanation: with only two packets, a single resolution failure collapses the entire duration to zero. This collapse propagates to every other feature that depends on the timestamp value, like `Flow IAT Mean`, `Flow IAT Max`, `Flow IAT Min`, and all rate-based features reduce to zero as well.

These rows impact is as follows. Globally, 353 unique row carry at least one NaN value making it 0.01% of the dataset, and 1,564 unique row carry at least one Inf value making it 0.05% of the dataset. At class level, Inf impact BENIGN with 1,427 unique row making it 0.06% of the class samples and only four attack classes are affected. NaN infects only BENIGN with 350 making it less then 0.01% of the class samples along with Dos Hulk. Rare classes such as Heartbleed, Infiltration, and Web Attack – SQL Injection are not affected. The attack class-level distribution is shown in the 4.7 Figure.

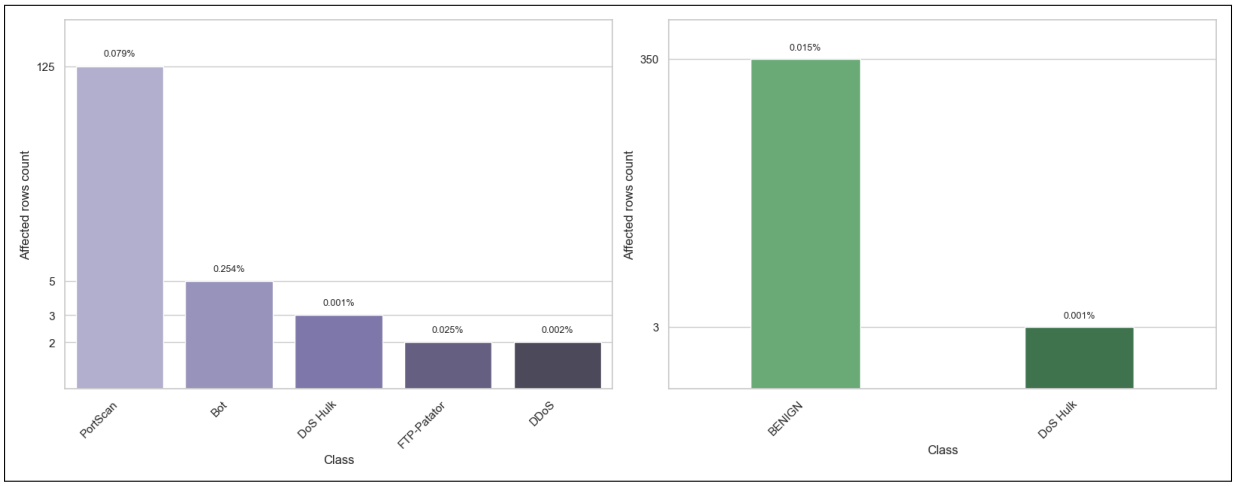


FIGURE 4.7 – Inf and NaN attack class distribution

Negative Values: Given the nature of the dataset feature, their logical value should never be negative yet Negative values were found across several feature groups, each with a distinct cause.

- **Sentinel encoding:** `Init_Win_bytes_forward` and `Init_Win_bytes_backward`.

  These two features carry negative values at scale: 1,001,189 rows (35.4%) and 1,441,552 rows (50.9%) respectively hold the value -1. This is not a measurement error. CICFlowMeter uses -1 as a sentinel to signal that the TCP initial window size could not be captured, either because the SYN packet was absent from the capture, or because the flow is UDP-based and has no TCP handshake. It is a deliberate encoding choice by the tool, not a data corruption.

- **Packet ordering artifacts:** `Flow Duration`, `Flow IAT Mean`, `Flow IAT Max`, `Flow IAT Min`, `Fwd IAT Min`, `Flow Bytes/s`, and `Flow Packets/s`.

  A total of 115 rows carry negative values in `Flow Duration` ranging from -13 to -1, with the same negative values propagating identically into `Flow IAT Mean`, `Flow IAT Max`, and `Flow IAT Min`.

  This is a cascade, since each of these rows has exactly one packet forward and one backward, there is only one inter-arrival interval in the flow, so mean, max, and min all equal `Flow Duration` exactly.

  Division by a small negative duration scaled by 1,000,000 then causes `Flow Bytes/s` and `Flow Packets/s` to carry large negative values.
  For example, with `Flow Duration` $= -1$ and 12 bytes of payload:

$$\texttt{Flow Bytes/s} = \frac{12}{-1/1{,}000{,}000} = -12{,}000{,}000$$

  This is likely to be a packet ordering artifact when processing the packets; additionally, CICFlowMeter performs no guard against negative duration, so the error propagates unchecked. The affected rows are exclusively BENIGN traffic and represent less than 0.0001% of the dataset.

  After excluding these 115 rows, an additional 2,776 rows exhibit negative `Flow IAT Min` values and 17 rows exhibit a negative `Fwd IAT Min`. Here the ordering artifact affects only a single packet pair within a longer multi-packet flow, dragging the minimum IAT negative while all other temporal features remain valid. The attack class-level distribution is shown in the 4.8 Figure.

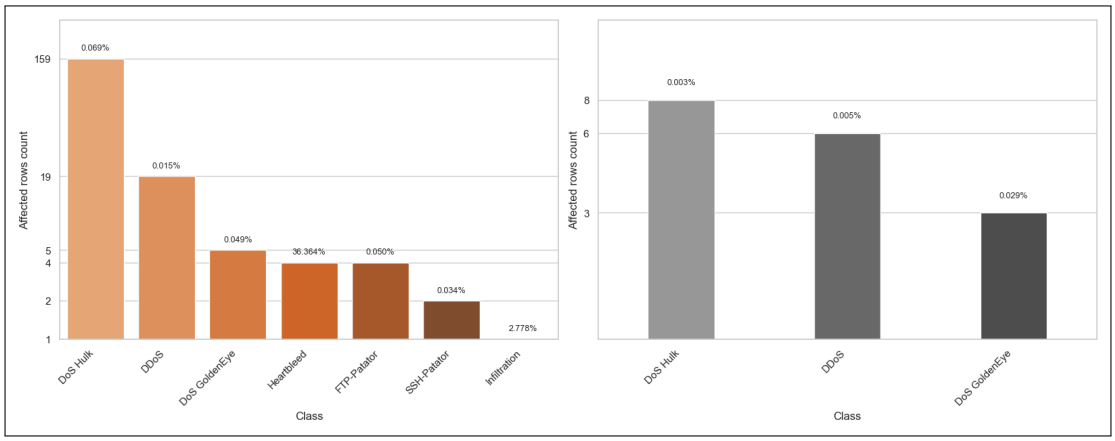


FIGURE 4.8 – Ordering Artifact Class Distribution

- **UDP parsing bug:** `Fwd Header Length`, `Fwd Header Length.1`, `Bwd Header Length`, `min_seg_size_forward`.

  These features exhibit values in the range of -32,212,234,632 to -1,073,741,320, far outside any legitimate segment size and a TCP header size, which is bounded between 20 and 60 bytes.

  The root cause is a bug in CICFlowMeter: when processing UDP packets, the tool reads the header size from a TCP-specific field that simply does not apply to UDP, producing meaningless values that corrupt these features.

The bug is confirmed by the data: all 35 affected rows are UDP flows, as indicated by `Init_Win_bytes_forward` = `Init_Win_bytes_backward` = -1.

The impact is narrow, 35 rows in `Fwd Header Length`, `Fwd Header Length.1`, and `min_seg_size_forward`, and 22 rows in `Bwd Header Length`, all exclusively BENIGN traffic, collectively under 0.0001% of the dataset.

Exact duplicates: Duplicate rows are common in network datasets since a row in the dataset does not represent a packet but a session behaviour summarised through statistical aggregation. Many flows share identical statistical profiles, thus the presence of duplicates does not necessarily indicate a data error.

Globally, duplicate rows represent 308,381 out of 2,830,743 rows (10.89%). Their distribution by attack class is as show in the 4.9 figure and no minority class is affected.

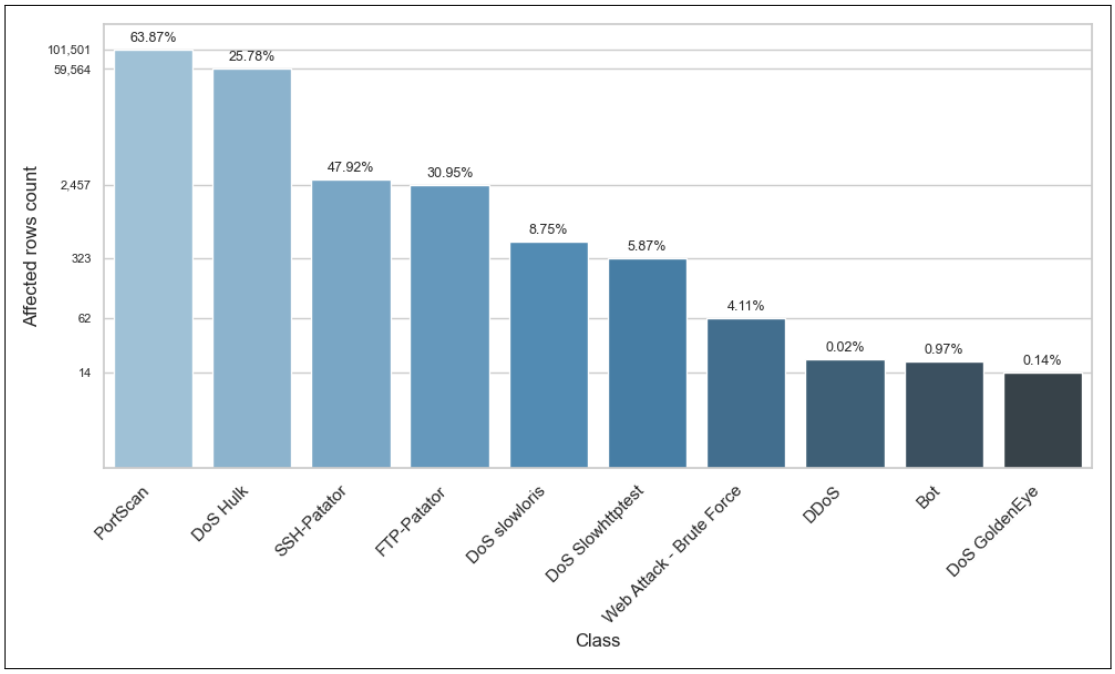


FIGURE 4.9 – Exact duplicates attack class distribution

Contradictory duplicates: These rows share the exact same feature values except for the target column, their labels differ. CICFlowMeter aggregates raw packet-level captures into flow statistics. Two flows from different sessions, one BENIGN and one malicious, can produce statistically identical feature vectors if their traffic patterns are indistinguishable at the flow-aggregation level. This is a fundamental limitation of statistical flow-based features rather than a labelling error.

Globally, 7,020 rows are affected. BENIGN traffic is the dominant participant in contradictory groups, as expected, BENIGN flows vastly outnumber attack flows, so the probability of a BENIGN flow sharing a feature vector with an attack flow is higher than inter-attack collisions. Contradictions involve only well-represented classes: DoS Hulk, DDoS, DoS slowloris, and PortScan. The 4.10 figure illustrates their distribution.

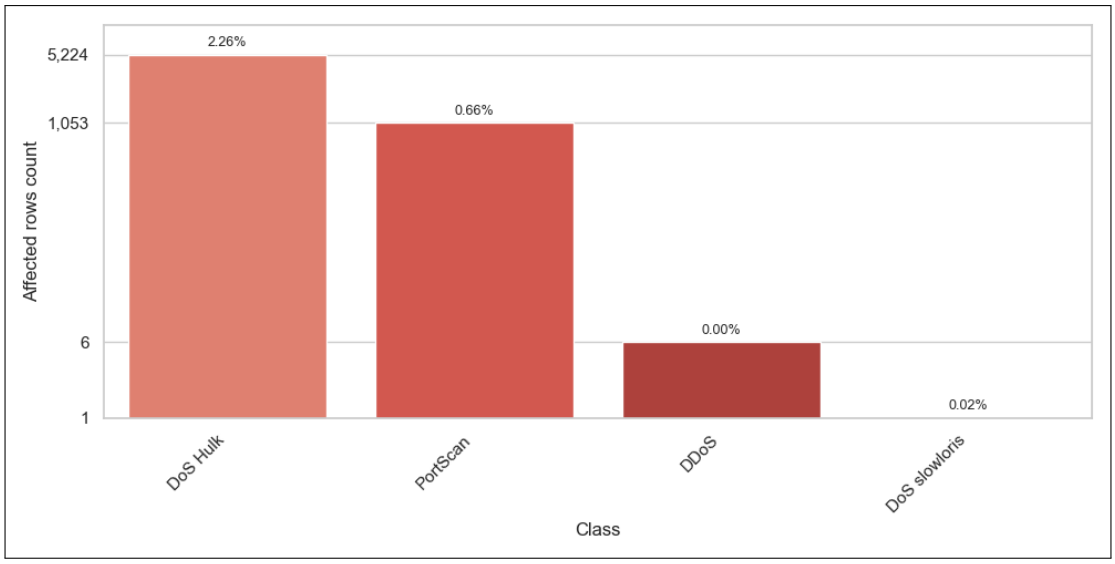


FIGURE 4.10 – Contradictory duplicates attack class distribution

Constant features: Constant features are those with a single unique value across the entire dataset, in this case, always zero. To understand this behaviour, each feature was traced back to its computation logic in the CICFlowMeter source code. The root cause divides them into two groups.

- **Bulk transfer features:** `Fwd Avg Bytes/Bulk`, `Fwd Avg Packets/Bulk`, `Fwd Avg Bulk Rate`, `Bwd Avg Bytes/Bulk`, `Bwd Avg Packets/Bulk`, `Bwd Avg Bulk Rate`.

  Bulk transfer refers to sending a large number of packets consecutively in one direction, with very little time between them.

  These features are derived from CICFlowMeter's bulk detection logic, which only activates when at least four consecutive packets carrying actual data are observed in close succession.

  If that condition is never met, either because flows in this dataset rarely involve this kind of continuous data sending, or because many packets carry no payload, the counters stay at zero and all six features are exported as zeros.

  This is exactly what is observed. These features are constant not by error, but because the bulk detection logic simply never triggers on this dataset.

- **Backward flag counters:** `Bwd PSH Flags` and `Bwd URG Flags`

  These are constant zero due to a bug in CICFlowMeter's implementation. The features `Bwd PSH Flags` and `Bwd URG Flags` are meant to count how many times the PSH and URG flags appear in packets traveling in the backward direction.

  However, these counters are only incremented inside the method that processes the first packet. Since the first packet by definition can only be forward, CICFlowMeter assigns the source IP from that first packet and compares subsequent packets against it to determine direction. The method that handles the following packets never updates these counters, so both features remain zero for every flow, regardless of

the actual traffic.

Near-constant features: `Fwd URG Flags`, `CWE Flag Count`, `RST Flag Count`, and `ECE Flag Count`.

Each feature hold exactly two unique values across the entire dataset. These are rare-event TCP control flags that are virtually absent in normal traffic. Combined, their non-zero occurrences account for only 315 rows (0.0139%), all within BENIGN traffic. Therfore, they carry no discriminative signal for any attack class.

4.4 Data preparatation

This section translates the findings from the data understanding phase into concrete preprocessing actions.

4.4.1 Data cleaning

The integrity of data is a prerequisite for any reliable model. Thus, we leverage data correctness over sample retention, as long as data volume and class balance are not meaningfully affected.

NaN and Inf Values: Rows carrying NaN or Inf values in `Flow Bytes/s` and `Flow Packets/s` are removed. As established, these arise from division by a zero `Flow Duration`, due to the lack of time granularity. Beyond the invalid rate features, these rows are also dominated by zeros which do not reflect the true nature of the flow, but are simply an artifact of insufficient measurement precision. Such a row cannot serve as a valid representation of a network session, as its defining characteristics were lost in the granularity gap.

The scale reinforces the decision. Globally, fewer than 0.11% of rows are affected. Additionally, the unique affected rows per attack class amount to 5 for Bot, 125 for PortScan, 2 for DDoS, and 3 for DoS Hulk, representing less than 0.00% of any single class, all of which carry sufficient clean samples to generalise from. No minority class is affected.

Negative Values: Negative values across the dataset fall into four distinct categories, each handled differently.

- **Sentinel encoding:** `Init_Win_bytes_forward` and `Init_Win_bytes_backward`.

  These are retained without modification. As noted, -1 is a deliberate sentinel used by CICFlowMeter to encode the absence of a TCP handshake, it is semantically valid and carries information about the flow type. Replacing it would discard a meaningful signal.

- **Packet ordering artifacts:** This anomaly originates during packet processing, since these sessions consist of a single packet exchange, the flow has no other packets to establish a valid duration or inter-arrival times. `Flow Duration` was rendered negative, and all rate-based features, `Flow IAT Mean`, `Flow IAT Max`, and `Flow IAT Min` collapsed to that same negative value, leaving no part of the flow intact.

Beyond the specific features where the corruption is visible, there is a deeper concern. This is a bug introduced during dataset construction, and its effects are not fully observable from the CSV alone.

Negative values in `Flow Duration` make the anomaly detectable, but there is no guarantee that the ordering artifact confined its damage only to features that produce obviously invalid numbers.

The same error may have silently distorted other features into values that appear plausible but are incorrect, damage that is invisible precisely because it does not manifest as negatives or zeros, but as ordinary-looking numbers that simply do not reflect the true flow.

In order to preserve data integrity and given the rows effect is negligible, only 115 rows belonging to BENIGN traffic, we decide to drop these rows.

`Flow IAT Min` and `Fwd IAT Min` share the same root cause, a packet ordering artifact, but the outcome is fundamentally different. Here the affected flows span multiple packets, so the ordering error disturbed only a single packet pair within an otherwise well-measured session.

The negative value appears only in the minimum IAT statistic, while `Flow Duration` remains positive and all other features are valid.

Inter-arrival time cannot be negative by definition, so clipping both features to zero restores the value to the physical boundary it should never have crossed, without fabricating anything.

This targeted correction is justified precisely because the flow itself was captured correctly, unlike the single-packet cases above, where the corruption was total and removal was the only option.

This treatment also preserves the 4 Heartbleed rows that would otherwise be discarded, a meaningful consideration given the class contains only 11 samples in total.

- **UDP parsing bug:** `Fwd Header Length`, `Bwd Header Length`, `Fwd Header Length.1`, and `min_seg_size_forward`.

  These rows are removed. As established, they are caused by a parsing bug rendering these feature values meaningless and irrecoverable. The fact that all affected rows belong exclusively to BENIGN traffic and represent under 0.002% of the dataset confirms that removal incurs no meaningful cost to class balance or dataset integrity.

Exact duplicates: Exact duplicate rows are dropped. Retaining them risks data leakage, a row could appear in both the training and test sets simultaneously, moreover it risks overfitting by biasing the model toward a single flow pattern. Since no minority classes are affected, the removal carries no risk to rare attack samples.

Contradictory duplicates: These rows present a serious risk to model training. An identical feature vector mapped to more than one target is unresolvable, retaining such rows forces the model to fit contradictory labels from the same input. All rows belonging

to contradictory groups are therefore dropped entirely, regardless of their label. Since rare classes are unaffected, this carries no risk of losing minority attack samples.

4.4.2 Feature engineering

The correlation matrix is recomputed on the cleaned data as the composition of the dataset has changed. 13 new pairs emerge and 3 pairs no longer appear, making a total of 49 pairs of highly correlated features, all of which provide no additional information to the model, some of which even have bugs in their computation causing the correlation.

The absence of the 3 pairs is explained by the cleaning process: `Fwd URG Flags` and `CWE Flag Count`, `RST Flag Count` and `ECE Flag Count` are near-constant features, and `Fwd Header Length` and `Fwd Header Length.1` are duplicate columns. As these features are no longer present in the dataset, the correlated pairs they formed no longer appear in the recomputed correlation analysis.

For the 13 new pairs, each pair contains at least one of the following features: `Fwd Header Length`, `Bwd Header Length`. These two features have their UDP parsing-corrupted rows dropped during cleaning. As a result, the true underlying correlation between these features and others becomes visible, previously masked by the corrupted values.

One feature from each pair is dropped. The choice is made based on Mutual Information (MI) score, which measures the feature-target association to determine which feature is more informative to distinguish between classes. For pairs with similar MI scores, the decision is made based on the domain knowledge.

Mutual Information measures how much knowing the value of a feature reduces uncertainty about the target class. Unlike the Pearson correlation, it captures both linear and non-linear relationships and works with categorical and continuous features. An MI score of 0 means it provides no useful information for target prediction, while a higher score indicates a stronger association with the target class. The MI score measures the association between a single feature and the target independently of other features:

$$MI(X;Y) = \sum_{x \in X} \sum_{y \in Y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \tag{4.2}$$

Where X is the feature, Y is the target, $p(x,y)$ is the joint probability distribution of X and Y, and $p(x)$ and $p(y)$ are the marginal probability of X and Y independently.

For 37 pairs of the 49 highly correlated pairs, the feature with the higher MI score is retained. However, these decisions may be overridden by the Knowledge-based decision.

For the remaining 12 pairs is made by examining what each feature represents in terms of network behavior rather than its statistical properties. The decision is made as follows for these pairs:

- **Packet and Segment length mean pairs:** `Fwd Packet Length Mean` and `Avg Fwd Segment Size`, `Bwd Packet Length Mean` and `Avg Bwd Segment Size`

- **Packet and Segment length mean pairs:** `Fwd Packet Length Mean` and `Avg Fwd Segment Size`, `Bwd Packet Length Mean` and `Avg Bwd Segment Size`, the

packets features are kept over the segments one, as they use standard network traffic analysis terminology, are immediately interpretable, and are consistent with the other packet length features already present in the dataset.

- **Totals and Subflows pairs:** `Total Fwd Packets` and `Subflow Fwd Packets`, `Total Backward Packets` and `Subflow Bwd Packets`, `Total Length of Fwd Packets` and `Subflow Fwd Bytes`, and `Total Length of Bwd Packets` and `Subflow Bwd Bytes`, as established before, the subflows are equal to the totals throughout the dataset due to a bug in their computation. The totals are kept as they are both correctly computed and more fundamental,representing the actual flow-level counts and byte totals directly. The subflow features are dropped as they provide no additional information beyond what the totals already capture.

- **Flags pair:** `Fwd PSH Flags` and `SYN Flag Count`, Although they are both only counted on the first packet, they still provide information about whether the flow was captured from the beginning of a TCP connection. `SYN Flag Count` is kept as it represents the synchronization packets counts used to initiate TCP connections making it more useful for that purpose, while the `Fwd PSH Flags` represents the count of how many forward packets had the push (PSH) flag set signaling the sender to push buffered data immediately.

- **Flags pair:** `Fwd PSH Flags` and `SYN Flag Count`, although both are only counted on the first packet due to the flag-counting bug, they still provide information about whether the flow was captured from the beginning of a TCP connection. `SYN Flag Count` is kept as it captures this condition more directly, reflecting whether a connection-initiating SYN packet was observed, which is more meaningful for distinguishing attack types than `Fwd PSH Flags`, which only indicates whether the first packet had the PSH flag set and carries no additional interpretable information beyond what `SYN Flag Count` already provides.

- **Duration and IAT pair:** `Flow Duration` and `Fwd IAT Total`, are highly correlated as both grow with the length of the connection, but they measure different quantities. `Flow Duration` captures the total elapsed time of the entire flow across both directions and is retained as it is a direct, universally understood measure of connection length. `Fwd IAT Total` measures the cumulative inter-arrival time between consecutive forward packets only, making it a direction-specific partial measure whose information is already partially represented by `Flow Duration` in combination with the other IAT features, and is therefore dropped.

- **Idle multicollinearity pairs:** `idle mean` and `idle max`, `idle mean` and `idle min`, `idle max` and `idle min`, the latter pair's features are both dropped in favor of `Idle Mean`, the two represent the extremes of the idle time distribution and are jointly collinear with it. `Idle Max` is susceptible to inflation by a single unusually long pause without reflecting the overall idle behavior of the flow, while `Idle Min` is easily influenced by noise from brief transient gaps. `Idle Mean` is retained as it is more representative of the typical idle behavior across the flow and less sensitive to individual outlier values.

In conclusion, of the 33 features involved in the highly correlated pairs, 19 are dropped:

- Subflow Bwd Bytes
- Subflow Bwd Packets
- Subflow Fwd Bytes
- Subflow Fwd Packets
- Avg Fwd Segment Size
- Bwd Header Length
- Bwd Packet Length Max
- Bwd Packet Length Std
- Flow IAT Max
- Fwd Header Length
- Fwd IAT Total
- Fwd PSH Flags
- Fwd Packet Length Std
- Fwd Packets/s
- Idle Max
- Idle Min
- Max Packet Length
- Packet Length Mean

4.4.3 Label encoding

The selected model, XGBoost, requires numerical input for both features and the target variable. While all features are already numerical, the target column `Label` is categorical and stored as an object type. Hence, Label Encoding was performed using Label encoder from sklearn.preprocessing. The encoder was fitted and applied to the target column in a single step using `fit_transform()`. To preserve interpretability of the model's predicted outputs, the resulting integer-to-class mapping was exported as a `label_mapping.json` file derived from `le.classes_`, allowing any predicted label to be traced back to its original class name.

4.4.4 Train/ Test split

Several factors make us approach this step with caution, when assessing the train/test split strategies.

A per-file split is not an option, as the files are divided per attack type, each attack class appears in exactly one file, so splitting by file would result in classes entirely absent from either the training or test set.

A standard random 80/20 split presents its own issues. The CIC-IDS2017 dataset is heavily imbalanced, the BENIGN to attack ratio is 80/20 Additionally, rare classes present less than 0.001% of the full dataset, so this strategy offers no guarantee of a fair class distribution and gives no assurance that rare classes will be adequately represented in both sets, which is particularly concerning given their scarcity.

For these reasons, a stratified train/test split is selected. It ensures all classes are present in both the training and test sets, each maintaining their original proportions, so the class distribution in each split mirrors that of the full dataset.

However, this strategy doesn't solve the class imbalance. This issue will be addressed and treated more during the model training.

4.5 Conclusion

This chapter presented the full data workflow adopted for this project, from selecting an appropriate public intrusion detection dataset to preparing it for model training. We compared candidate datasets and justified the choice of CIC-IDS2017, then analyzed data quality issues such as NaN/Inf values, negative artifacts, duplicates, and constant features, linking each issue to its root cause in traffic generation or feature extraction. Finally, we applied cleaning and preparation steps that preserve class coverage and prevent leakage, and we defined an encoding and stratified splitting strategy to support reliable evaluation in the subsequent modeling chapter.

Chapter 5

Modeling

5.1 Introduction

5.2 State of the Art

5.3 Adopted Approach

5.4 Training and Hyperparameter Optimization

This section details the full training and optimization process of the XGBoost classifier on the CIC-IDS2017 dataset.

5.4.1 Initial Setup

Metric selection :

The CIC-IDS2017 dataset is inherently imbalanced. BENIGN traffic constitutes roughly 80% of the dataset, reflecting the reality of real network traffic where attacks are rare events. Within the attack classes themselves, representation varies significantly, some classes contain hundreds of thousands of samples while others hold fewer than a dozen. This makes metric selection a non-trivial decision that directly affects how honestly model performance is reported.

Accuracy is an intuitive metric but an unreliable one here. It counts correct predictions across all samples divided by the total:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

The sheer volume of BENIGN instances inflates the score regardless of attack detection quality. A model that predicts BENIGN for every single sample would still achieve over

99% accuracy while detecting zero attacks. T his makes accuracy effectively blind to what actually matters.

Precision and recall offer more granular insight into model behavior. Precision measures how many of the model's predictions for a given class are actually correct, and Recall measures how many of the true instances of that class the model successfully detected:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Both metrics are informative individually but insufficient alone; a model biased toward conservative predictions will report high precision at the cost of missing many attacks, while an overly permissive model will recover most attacks but introduce a high rate of false positives.

The F1-score addresses this by taking the harmonic mean of precision and recall, penalizing models that sacrifice one for the other:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

However, standard F1-score or its weighted variant still allows dominant classes to overshadow rare ones in the aggregate score, since each class's contribution to the final score is proportional to its sample count.

We therefore adopt the Macro F1-score as the primary evaluation metric, which computes F1-score separately for each class, then averages those scores giving every class the same contribution regardless of how many samples it contains. It is defined as:

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} \text{F1}_i$$

where C is the total number of classes and F1_i is the F1-score for class i.

This guarantees that every class, regardless of size, contributes equally to the final score. Per-class F1-score scores are also tracked alongside it throughout all experiments to monitor individual class behavior.

Cross-Validation :

Given the challenges associated with the CIC-IDS2017 dataset, particularly the class imbalance issue and the limited number of samples in certain attack classes, we decide to adopt an experimental approach during the training phase and rely on evaluation metrics to guide our decisions for the final model configuration. This led us to use cross validation.

Cross-validation is an evaluation technique that gives a more stable and reliable estimate of model performance, instead of relying on a single train-test split, the dataset is divided into several folds, and the model is trained and evaluated multiple times, each time using a different fold as the validation set while the remaining folds are used for training. The final performance is then obtained by averaging the results across all runs. Several variants exist:

- Standard k-fold divides the data into k equal folds and rotates the validation fold across all k iterations. It is more practical, but it assigns samples to folds randomly, with no guarantee that rare classes appear in every fold. With classes as scarce as Heartbleed, a random split could place all its samples in the training folds and leave none for validation, or the reverse, making the evaluation meaningless for that class.

- Stratified k-fold addresses this directly. It applies the same k-fold rotation but enforces that each fold mirrors the class distribution of the full training set. If Heartbleed represents 0.0005% of the data, each fold will contain approximately that proportion. This guarantees that every class is present in both the training and validation portions of every fold, producing evaluation scores that are consistent and comparable across all k runs. We set k = 5. This choice is driven primarily by the scarcity of our rarest class: with only 9 Heartbleed samples in the training set, each fold sees approximately 2 instances. Increasing k would reduce this further, making per-fold evaluation for rare classes even less stable. Decreasing it would produce fewer evaluation rounds and increase variance in the aggregated score. Five folds represent a practical balance between evaluation stability and computational cost given this constraint.

Across all experiments, the reported performance corresponds to the mean Macro F1-score over the five folds, with per-class F1-score scores tracked individually to monitor class-level behavior throughout.

Initial XGBoost parameters

5.4.2 Baseline training

A baseline model should be the first model built before moving toward more complex approaches, as it acts as a reference point for evaluating the performance of future configurations.

If a more advanced configuration fails to outperform the baseline or produces even lower results, this may indicate that the additional complexity provides little practical benefit. In such cases, the issue may not originate from the model itself, but may instead suggest potential limitations related to the dataset.

Furthermore, baseline results provide guidance for future steps. For example, if a classification model fails to predict certain classes, efforts should be placed on addressing this flaw.

For these reasons, we establish a baseline model by training XGBoost. The baseline training results divide into expected and unexpected outcomes, as shown in Figure 5.1:

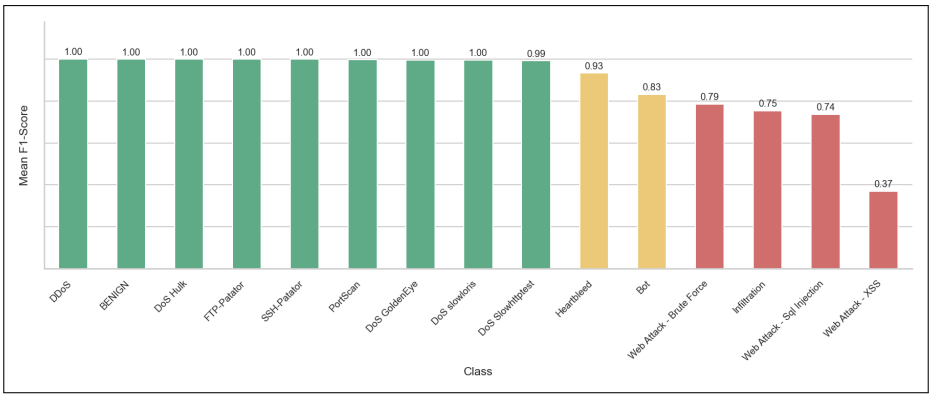


FIGURE 5.1 – Baseline F1-score per class

Expected outcomes :

Classes with large sample counts perform exactly as anticipated. Attack types such as DoS Hulk, PortScan, and DDoS, each with over 100,000 training samples, along with moderately represented classes like DoS GoldenEye, FTP-Patator, SSH-Patator, DoS Slowloris, and DoS Slowhttptest, each above 5,000 samples, achieve a high F1-score at 0.99. The model has enough examples to learn stable, discriminative boundaries for these classes.

At the other end of the spectrum, Heartbleed and Infiltration show lower performance, with F1-score at 0.93 and 0.75 respectively. With only 29 and 9 training samples remaining after preprocessing, this outcome is fully expected, there is simply not enough signal for the model to generalize reliably.

Unexpected outcomes :

- **Web Attacks:** The more revealing result comes from the Web attack subclasses. We anticipated that Web Attack – SQL Injection would struggle, given only 19 training samples. What was unexpected is that all three Web Attack subclasses perform poorly, Brute Force achieves F1-score = 0.78, SQL Injection F1-score = 0.74, and XSS a striking F1-score = 0.37, despite having 522 training samples, supposedly sufficient for a tree-based model to learn from.

  The confusion matrix is computed to identify the specific misclassification patterns. As shown in figure 5.2, there is a consistent mutual confusion between XSS and Brute Force, XSS is misclassified 345 times as Web Attack – Brute Force, and Web Attack – Brute Force is misclassified as XSS 167 times. Both are also occasionally classified as BENIGN. This indicates that the model fails to establish meaningful separability between these categories in the feature space.

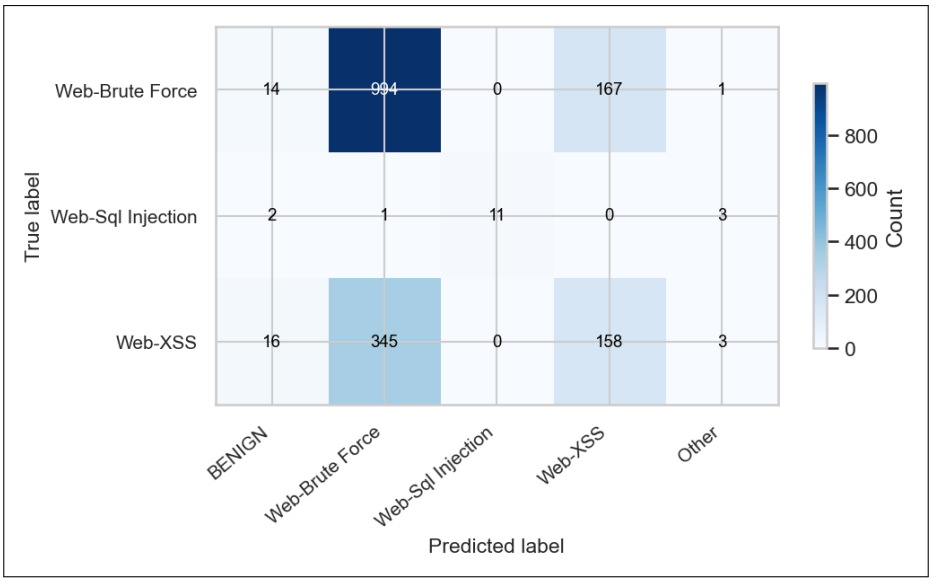


FIGURE 5.2 – Web Attack baseline confusion matrix

Although the Web Attack subclasses remain relatively underrepresented compared to dominant classes in the dataset their consistently poor performance suggests something potentially more fundamental than class imbalance or sample scarcity alone.

Web attacks are payload-based attacks, an XSS carries a malicious script in a HTTP request body, an SQL injection craftes SQL commands by tempering with input fields interfering with database queries. Flow-based datasets like CIC-IDS2017 record features derived from packet timing, length distributions, and header-level metadata; there is no payload field. These features are evidently insufficient to capture the semantic structure of such attacks.

- **Bot attack:** Bot is still relatively underrepresented compared to dominant classes, however, with approximately 1,500 training samples, it was initially expected to provide sufficient information for the model to learn meaningful patterns. Despite this its performance remains poor with only an F1-score at 0.82.

  Considering the attack's nature, might explain why the model is failing it.

  A Bot attack typically operates in multiple phases. The first is the infection phase, often done through phishing, or malicious download. It occurs without any visible indications, and the user remains unaware that their machine is under the control of an attacker, effectively turning it into a "zombie" machine. The second is Command-and-Control (C2) communication where the infected machine establishes contact with an external server controlled by the attacker to receive instructions and periodically report its status. And finally using the botnet to launch an attack such as DDoS attacks, spam distribution, or data exfiltration. CIC-IDS2017 represents the second phase.

  The confusion matrix in Figure 5.3 shows that bot attack is being misclassified 351 times as benign. This is due to the fact that, in this phase, the attacker

intentionally mimics benign traffic patterns in order to remain undetected, making it difficult for the model to distinguish between malicious and normal behavior. Moreover, since the dataset represents each flow independently, it fails to capture the temporal nature of bot behavior, particularly periodic communication patterns across sessions.

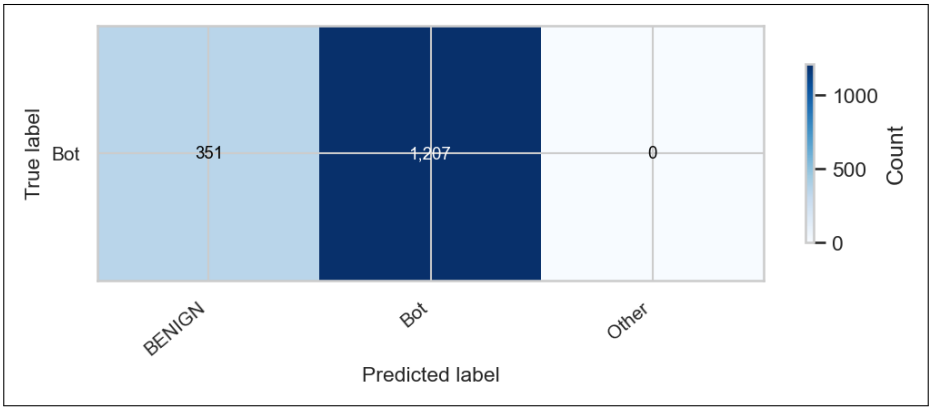


FIGURE 5.3 – Bot attack baseline confusion matrix

The baseline results show that underperforming classes do not fail for a single unified reason. Instead, two different patterns emerge. On one hand, classes such as Heartbleed and Infiltration appear to be primarily affected by extreme sample scarcity. On the other hand, Bot and Web Attack subclasses exhibit behavior that suggests additional limitations related to the nature of the data representation itself.

Despite this distinction, all these classes are still, to varying degrees, underrepresented compared to dominant classes. For this reason, it remains reasonable to investigate whether standard imbalance mitigation techniques can improve their performance before drawing definitive conclusions about the nature of these limitations.

5.4.3 Iterative Experiments

The class imbalance problem can be addressed at two levels with either an algorithm-level approach or a data-level approach :

- **Algorithm-level: Class Weights**

  XGBoost is a gradient boosting model that learns by minimizing a loss function, for every wrongly predicted sample, it multiplies the loss associated with it by its assigned weight which is one by default, meaning the sample count is what primarily drives the loss function.

  In highly imbalanced datasets such as CIC-IDS2017, the model naturally optimizes for the majority class because of their higher sample count.

  In XGBoost, sample weights are used to adjust the importance of each individual training instance in the loss function, effectively penalizing the model more heavily for misclassifying samples. By assigning higher weights to the minority class,

this forces the algorithm to focus on correctly predicting those rare instances in subsequent boosting rounds.

This technique is a standard way to handle imbalanced datasets without modifying the training data.

- **Data-level: Data augmentation**

  SMOTE (Synthetic Minority Oversampling Technique) generates new synthetic samples rather than duplicating existing ones. SMOTE works by identifying the k-nearest neighbors of an instance then selects one at random.

  It then generates a new synthetic point using linear interpolation; it computes the distance between the two points, multiples the difference by a random number between 0 and 1 and adds this value to the original sample creating the artificial new sample.

  Although SMOTE becomes risky when applied to classes with very low sample counts, as the synthetic samples are created from a very limited neighborhood, which reduces diversity and can lead to unrealistic decision boundaries. This may result in generating synthetic points that do not accurately represent the real characteristics of that class, and can introduce noise.

  In this case Random over sampling (ROS) becomes a better resort. It works by duplicating existing samples. It introduces no new information but ensures the class appears more frequently during training.

That said, we approach SMOTE with deliberate caution. Network intrusion data is sensitive. A synthetic sample generated by interpolating between two real flow records is not guaranteed to faithfully capture the attack pattern, which presents a real risk. It may introduce noise that disrupts the model's learned decision boundaries rather than reinforcing them. Given the critical nature of network intrusion detection, we favor XGBoost's class weighting over data augmentation, as it offers a simpler and safer mechanism that achieves a similar effect without modifying the training data. Nonetheless, we still run sampling experiments to evaluate whether the data-level approach provides any measurable gain.

We define a fixed set of experiments. If a new configuration involves altering the training set, the same experiments are rerun to ensure consistent evaluation:

1. Class weights
2. Sampling approaches (ROS and SMOTE)
3. SMOTE combined with weights

Following baseline training, we attempt to improve the performance of weaker-performing classes, namely the Web Attack subclasses, Heartbleed, Infiltration, and Bot, through these series of experiments while monitoring the evaluation metrics.

Under class weighting (balanced), Infiltration improves to an F1-score of 0.8685 and Heartbleed reaches a perfect score of 1.0. However, Bot performance deteriorates, while Web Attack classes show no measurable improvement.

Alternative Weight Formulas: A second weighting strategy is explored using two custom formulations. The first assigns weights inversely proportional to the square of the baseline F1-score, amplifying attention on the worst-performing classes disproportionately more. The second is a hybrid formula that combines a frequency-based term, the square root of the inverse class frequency, with an F1-based penalty, the inverse of the baseline F1-score, so classes that are both rare and poorly predicted receive the highest combined weight. Both formulas are normalized by their mean to keep training numerically stable. However, neither outperforms the standard balanced weighting; no improvement is observed for the target classes, and this configuration is dropped from all subsequent experiment sets.

Sampling (SMOTE + Random Oversampling):

Sampling is applied only within the training folds during cross-validation. The resampling is fit exclusively on each training fold, never on the validation fold, to prevent any form of data leakage from the synthetic samples into the evaluation.

In SMOTE, the parameter k defines the number of nearest neighbors used to generate synthetic samples. We set k = 5, the default value, as it ensures a stable neighborhood structure for synthetic sample generation. This choice directly influences the minimum viable class size required for SMOTE to operate correctly.

Based on this constraint, we apply SMOTE only to classes with more than 50 training samples and ROS on the ones below. This threshold is not arbitrary, SMOTE requires at least k + 1 samples in the minority class to identify valid neighbors. Below that count, the neighborhood becomes degenerate; the algorithm would interpolate between the same few points repeatedly, producing synthetic data that is essentially noise rather than new information.

The resampling targets are set as follows: SMOTE is used to increase Bot samples from 390 to 3,000, Web Attack – Brute Force from 294 to 3,000, and Web Attack – XSS from 522 to 2,000. Random oversampling is applied to Web Attack – SQL Injection from 17 to 300, Heartbleed from 9 to 300, and Infiltration from 29 to 300.

Under this configuration, Web Attack classes show no improvement. While Infiltration drops slightly compared to the weights configuration with F1-score = 0.8218, and Heartbleed remains at 1.0.

When combining sampling with class weighting, Bot performance drops further, whereas Infiltration recovers to F1-score = 0.9055. Web Attack classes remain unchanged.

The consistent finding across all four configurations is that Web Attack performance does not respond to any rebalancing approach. This confirms that this is not a class imbalance problem, it is a feature representation problem.

Relative to the baseline, Bot performance consistently drops under these configurations, which confirms that this is a dataset limitation.

Web Attack Class Merging

As established, no combination of weighting or sampling improves Web Attack performance. The confusion matrix shows persistent mutual misclassification between XSS and Brute Force across every experiment. Their flow-level signatures are too similar to

separate with the available features.

We therefore decide to merge the three Web Attack subclasses Brute Force, SQL Injection and XSS into a single unified Web Attack class.

Because this modifies the training set, all four experimental configurations are re-run from scratch.

The merged Web Attack class immediately achieves F1-score = 0.987. However, Infiltration drops from 0.7867 to 0.7267 and Heartbleed drops from 0.9333 to 0.7333 relative to the original baseline, while Bot remains stable at 0.83.

When class weights are introduced, Heartbleed returns to F1-score = 1.0. Infiltration recovers to 0.8685, matching its pre-merge weighted performance. Web Attack holds at 0.98.

Under sampling alone, Infiltration reaches only 0.74, Heartbleed remains at 1.0. Web Attack stays at 0.98.

Finally, combining sampling with class weights yields similar behavior, heartbleed and Web Attack stays at 1.0 and 0.98 respectively. However, infiltration returns to 0.83.

Across all four configurations, Web Attack consistently sits at approximately 0.98 regardless of whether weights or sampling are added. This confirms that merging alone is the decisive intervention; no additional rebalancing is necessary.

Heartbleed

Heartbleed warrants a closer look. At baseline, Heartbleed achieves an F1-score of 0.93, yet under every subsequent configuration, weights, sampling and a mix of both, it reliably reaches 1.0.

With only 9 training samples and 2 retained in the test set, cross-validation folds see approximately 2 Heartbleed samples per validation split. At that scale, the F1-score can take only a handful of discrete values, 0 when both are wrong, 0.67 in case only one is wrong, and a perfect 1.0 score means the model simply predicts both samples correctly.

More importantly, all resampling and rebalancing techniques are applied exclusively to the training folds and therefore do not affect the test distribution. As a result, regardless of whether weights or sampling are used, the final evaluation is always based on the same two test samples. This makes it hard to meaningfully assess the impact of these strategies on Heartbleed's generalization performance.

This initially motivates an ablation experiment in which Heartbleed is temporarily removed to assess its impact on the model's behavior.

Heartbleed ablation results

Following this change, the complete set of experiments is rerun. However, the results are unexpected: Infiltration, which previously showed significant improvement under the merged Web Attack configuration with class weighting and Heartbleed included, drops when Heartbleed is removed while keeping the same merge and weighting setup.

To verify whether this dependency is real or an artifact, we run a systematic comparison across three settings: pre-merge, merge with Heartbleed, and merge without

Heartbleed.

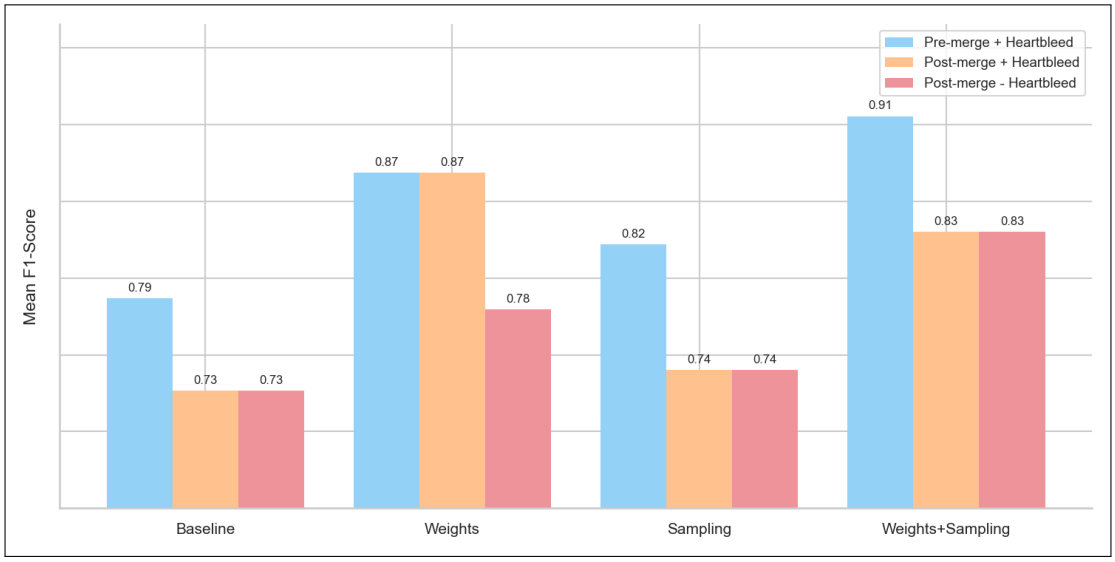


FIGURE 5.4 – Infiltration F1-score comparison across configurations

As shown in Figure 5.4, the Infiltration F1-score varies across the three configurations.

Merging Web Attacks consistently reduces Infiltration F1-score by 0.04 to 0.08 across all four strategies. This is likely because consolidating three Web Attack subtypes into one class reshapes the global decision boundary landscape in a way that slightly disadvantages Infiltration's region.

Dropping Heartbleed at baseline has zero effect on Infiltration, with identical scores ($0.7267 = 0.7267$).

Dropping Heartbleed under weighting reduces Infiltration F1-score by up to 0.089.

To further investigate this behavior, we construct a simplified 2D representation of the problem by training a separate XGBoost model using only two selected features. In reality, the original model operates in a high-dimensional feature space composed of 46 features, making direct visualization of its decision boundaries impossible. Reducing the problem to two dimensions allows the learned feature space partitioning to be visualized and interpreted more clearly.

To select these features, we first apply SHAP (SHapley Additive exPlanations) analysis independently on both Heartbleed and Infiltration. As shown in the SHAP value distribution figures, the two features contributing most significantly to the model's decisions for both classes are Total Length of Bwd Packets and Fwd Packet Length Max.

In the unweighted configuration with Heartbleed, as shown in Figure, the feature space is entirely dominated by the BENIGN class. Heartbleed samples cluster in the bottom-left corner but receive no dedicated boundary region. The boundary structure is flat and undifferentiated.

A similar behavior is observed in the unweighted configuration without Heartbleed, presented in Figure, the boundary is visually identical to the previous configuration. Without weighting, Heartbleed's presence or absence has no effect on the optimization landscape.

In contrast, the weighted configuration with Heartbleed, illustrated in Figure, the boundary structure changes substantially. A dedicated region is carved out around the Heartbleed samples along the Fwd Packet Length Max axis, and a series of sharp horizontal cuts appear across the feature space. These cuts reshape the decision surface in a way that incidentally improves the separation of Infiltration samples.

When Heartbleed is removed under weighting, as shown in Figure, the horizontal structure disappears entirely and the boundary reverts to the flat BENIGN-dominated surface observed in the unweighted cases.

This behavior can be explained by the optimization mechanism of XGBoost. Under class weighting, Heartbleed receives a gradient scaling factor of approximately 19,000× relative to the BENIGN class. Each of its 9 training samples exerts as much optimization pressure on the model as a substantial fraction of the majority class flows. The model, responding to this extreme gradient signal, prioritizes correctly resolving the Heartbleed decision boundary.

This extreme weighting has an indirect structural effect on Infiltration. In XGBoost, each tree is built by recursively partitioning the feature space, and each split is chosen to minimize the weighted gradient loss across all classes simultaneously. When Heartbleed receives 19,000× weight, its samples impose disproportionate pressure on how certain boundaries are drawn in the shared feature subspace.

Some of these boundaries, though oriented toward separating Heartbleed from BENIGN, incidentally create a favorable partition for Infiltration samples as well, because the two classes occupy adjacent regions of feature space, the cut drawn to isolate Heartbleed simultaneously pulls Infiltration away from BENIGN. Removing Heartbleed under weighting relaxes that boundary, and Infiltration loses the incidental separation from BENIGN.

To conclude, Heartbleed and Infiltration have no structural relationship in the data. The dependency is induced entirely by the extreme weight assigned to Heartbleed.

The ablation results demonstrate that removing or amplifying Heartbleed affects specific aspects of the model's behavior; under class weighting, its strong gradient signal contributes to sharpening nearby decision boundaries in the feature space, which indirectly benefits Infiltration performance.

This observation led to a reassessment of the initial interpretation of metric limitation.

As established, while the model achieves a perfect F1-score of 1.0, this is simply because it is evaluated on only two samples, which limits the metric to a small set of discrete outcomes, rendering evaluation unreliable. As a result, performance on this class cannot be reliably assessed from the evaluation set alone.

However, this limitation does not imply that the model fails to learn the class during training. The training process still incorporates Heartbleed samples, allowing the model to adjust its internal decision boundaries accordingly, even if this cannot be properly reflected in the test evaluation.

In light of the ablation findings, we decide to retain Heartbleed.

5.4.4 Final Model Configuration

The final model configuration carried forward into hyperparameter tuning consists of Web Attack subclasses merged into one class, Heartbleed retained, and the use of balanced class weights. Thus achieves a Macro F1-score at 0.98, with Infiltration F1-score = 0.8236, and Web Attack F1-score = 0.9896.

5.4.5 Hyper parameter tuning

5.4.6 Adding thresholds

5.4.7 Final model training results

5.5 Conclusion

Chapter 6

Realization

6.1 Introduction

«««¡ Updated upstream

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6.2 Integration

6.3 Deployment Diagram

A deployment diagram models the physical architecture of a system and illustrates how software components are deployed on hardware nodes[5]. Figure 6.1 presents the deployment diagram of our system.

FIGURE 6.1 – Deployment Diagram

»»»¿ Stashed changes

6.4 Development Tools and Technologies

TABLE 6.1 – Development tools and technologies

Language
 **Python**

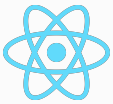
Machine Learning

Scikit-learn

Backend

Spring Boot

Frontend



React

Database

MySQL

Local Server

XAMPP

API Testing

Postman

API

FastAPI

Containerization

Docker

6.5 Performance Analysis and Evaluation Metrics

6.6 User Interfaces

This section presents the main user interfaces of the NIDS Web Application.

6.6.1 Use case «Authenticate» realization

The authentication interface allows the user to log in and access the platform features, as shown in Figure 6.2.

Placeholder screenshot for the **Authenticate** use case.

FIGURE 6.2 – Authentication interface (placeholder)

6.6.2 Use case «View Dashboard» realization

The dashboard interface provides a global overview of the system state and recent activity, as shown in Figure 6.3.

Placeholder screenshot for the **View Dashboard** use case.

FIGURE 6.3 – Dashboard interface (placeholder)

6.6.3 Use case «View Statistics Summary» realization

The statistics summary interface provides aggregated metrics and trends for monitoring purposes, as shown in Figure 6.4.

Placeholder screenshot for the **View Statistics Summary** use case.

FIGURE 6.4 – Statistics summary interface (placeholder)

6.6.4 Use case «View Alerts List» realization

The alerts list interface displays all detected alerts with key information for quick review, as shown in Figure 6.5.

Placeholder screenshot for the **View Alerts List** use case.

FIGURE 6.5 – Alerts list interface (placeholder)

6.6.5 Use case «View Alert Detail» realization

The alert detail interface presents the full information of a selected alert, enabling deeper investigation, as shown in Figure 6.6.

6.6.6 Use case «Search Alerts» realization

The search interface enables filtering and searching through alerts by different criteria, as shown in Figure 6.7.

6.6.7 Use case «Manage Alert Status» realization

The alert status management interface allows updating the handling state of an alert (e.g., new, in progress, resolved), as shown in Figure 6.8.

Placeholder screenshot for the **View Alert Detail** use case.

FIGURE 6.6 – Alert detail interface (placeholder)

Placeholder screenshot for the **Search Alerts** use case.

FIGURE 6.7 – Search alerts interface (placeholder)

6.6.8 Use case «Trigger Network Analysis» realization

The analysis trigger interface allows an administrator to start a network analysis process, as shown in Figure 6.9.

6.6.9 Use case «Analyze Network Flow» realization

The network flow analysis interface displays analysis outputs and indicators extracted from network traffic, as shown in Figure 6.10.

6.6.10 Use case «send Alert» realization

The alert sending interface triggers an alert notification to external services (e.g., email), as shown in Figure 6.11.

6.6.11 Use case «View Reports» realization

The reports interface allows the user to browse and consult generated reports, as shown in Figure 6.12.

6.6.12 Use case «Export Report» realization

The export interface enables downloading a report in a suitable format for sharing or archiving, as shown in Figure 6.13.

6.7 Integration

6.8 Conclusion

Placeholder screenshot for the **Manage Alert Status** use case.

FIGURE 6.8 – Manage alert status interface (placeholder)

Placeholder screenshot for the **Trigger Network Analysis** use case.

FIGURE 6.9 – Trigger network analysis interface (placeholder)

Placeholder screenshot for the **Analyze Network Flow** use case.

FIGURE 6.10 – Analyze network flow interface (placeholder)

Placeholder screenshot for the **send Alert** use case.

FIGURE 6.11 – Send alert interface (placeholder)

Placeholder screenshot for the **View Reports** use case.

FIGURE 6.12 – Reports interface (placeholder)

Placeholder screenshot for the **Export Report** use case.

FIGURE 6.13 – Export report interface (placeholder)

General Conclusion

Webography

[1] https://www.snort.org/. Accessed on: April 6, 2026.

[2] https://suricata.io/. Accessed on: April 6, 2026.

[3] https://www.ibm.com/docs/en/rsas/7.5.0?topic=structure-class-diagrams. Accessed on: April 4, 2026.

[4] https://github.com/ahlashkari/CICFlowMeter. Accessed on: April 23, 2026.

[5] https://www.ibm.com/. Accessed on: May 13, 2026.

Bibliography

[1] Chris Jackson. *Network Security Auditing*. 800 East 96th Street, Indianapolis, IN 46240 USA: Cisco Press, 2010. ISBN: 978-1-58705-352-8.

[2] R. Wirth and Jochen Hipp. "CRISP-DM: Towards a standard process model for data mining". In: *Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining* (Jan. 2000), pp. 4–7.

[3] John Erickson and Keng Siau. "Unified Modeling Language: The Teen Years and Growing Pains". In: vol. 8016. July 2013, pp. 3–6. ISBN: 978-3-642-39208-5. DOI: 10.1007/978-3-642-39209-2_34.

[4] Mahbod Tavallaee et al. "A Detailed Analysis of the KDD CUP 99 Data Set". In: *2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*. 2009, pp. 1–6. DOI: 10.1109/CISDA.2009.5356528.

[5] Nour Moustafa and Jill Slay. "UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)". In: *Military Communications and Information Systems Conference (MilCIS)*. IEEE, 2015. DOI: 10.1109/MilCIS.2015.7348942.

[6] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A Ghorbani. "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization". In: *ICISSP* (2018), pp. 108–116. DOI: 10.5220/0006639801080116.